

Dynamics of Self-Consistent Relational-Wave Closed Systems——From the local instability of the 90-degree relative equilibrium to inflation, spontaneous symmetry breaking, Grassmann geometry, terminal universality, and relationalism

Noriaki Kihara (WF System Co., Ltd.), ORCID 0009-0004-6753-4020

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Author: Noriaki Kihara (WF System Co., Ltd.) **Date:** 2026-09-12

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Abstract

For the dynamics defined in the prior work (Mechanism of inflationary rapid expansion in self-consistent relational-wave closed systems, Concept DOI 10.5281/zenodo.22112008)——place a complex relational wave z_e on each edge of the complete graph K_N , and drive it with the real orthogonal map $z \rightarrow \exp(\Delta\tau K)z$ ($\Delta\tau = 2\pi/N$) generated by the phase-only real antisymmetric generator $K_{ef} = A_{ef} \sin(\varphi_f - \varphi_e)$ (A is the adjacency of the line graph $L(K_N)$)——this capstone establishes the nine foundational papers (Papers 1–9) as a **single causal chain**:

map $\rightarrow$ 90-degree self-consistent relative equilibrium (floor) $\rightarrow$ ignition qualification = local instability of the floor $\rightarrow$

Central proposition: the local instability of the self-consistent 90-degree relative equilibrium produces **inflationary rapid expansion** (exponential growth of the out-of-floor-plane component $H_\perp/H$), spontaneous symmetry breaking, and plane reorientation; after nonlinear saturation it universalizes each wave's amplitude to $\sqrt{H/M}$ while leaving only the relational phase/geometric information. The **mechanism** of the rapid expansion is the linear instability of the floor, and its **real amplification rate** is refined by the finite-time Lyapunov rate of the time-varying tangent cocycle along the real orbit (phenomenon = inflation, mechanism = linear instability + time-varying tangent cocycle).

Each stage of the chain: (1) exact derivation of the single-wave equation of motion (Kuramoto-type phase coupling; amplitude driven by the second harmonic $\sin 2\Delta\varphi$; Paper 1). (2) On the 90-degree skeleton, $D^{-1}KD = iW$ ($D = \text{diag}(e^{i\varphi_e})$, W real symmetric nonnegative), so the complex self-consistent floor problem is exactly equivalent to the Perron problem of the real symmetric W ; a positive-amplitude floor exists uniquely, and from the parity bipartite structure σ^2 , each edge amplitude², and the full spectrum can be written in closed form for **arbitrary** N , with active dimension $O(N)$ (Papers 1 and 3). (3) The ignition qualification is “an unstable relative equilibrium” only; high symmetry, zero centroid, and square-closure are not qualifications (Paper 2). (4) **On the cyclic-Fourier high-symmetry floor** the factorization of zero-closure is quantized into discrete block sizes with $L = N/\text{gcd}(N, 4)$, $p = \text{spf}(L)$, and $p = 2 \iff 8 \mid N$ ($\nu_2(N) \geq 3$) is the 2-adic selection rule of the square map (Paper 4). (5) An infinitesimal seed breaks the symmetry and triggers **inflationary rapid expansion**. It begins with the floor-Jacobian initial/frozen rate $\lambda_0 = \ln \rho_0$ ($\rho_0 = \rho(J_0)$, = the Landau summit curvature $V''(0) = -\lambda_0$), and its **real amplification rate is the finite-time Lyapunov rate γ_t of the tangent cocycle $\Phi(t, 0) = \prod_t J(z_t)$ along the real orbit** (for $N = 6, 8, 10$, $\gamma_t > \lambda_0$) (Papers 5 and 6). (6) The state is a rank-2 plane $P(t)$ at each instant ($\Pi_{t+1} = Q_t \Pi_t Q_t^T$), with $H_\perp/H = \frac{1}{4}\|\Pi_t - \Pi_0\|_F^2$ —inflation is exponential departure from the initial plane on $\text{Gr}(2, M)$, and the terminal plane spans four dimensions with the original plane via two nonzero principal angles (the cumulative map is not $SO(4)$; Paper 7). (7) The terminus universalizes to equal amplitude $a_\infty = \sqrt{H/M}$ (a state quantity, extensive (H, M) / intensive $\rho = H/M$) and $\Delta_{\text{tail}} = -2\pi/N$ (Papers 5 and 8). The **degenerate vacuum family** is not identifiable as a finite discrete orbit of S_N + global phase + conjugation (it was confirmed that the five terminal states for $N = 6$ do not map to one another); the orientation is seed-dependent (SSB confirmed in all 35 values of $N \times 5$ seeds = 175 runs). That it is not a finite discrete orbit is established; continuity and the identification of a larger continuous symmetry group are open (Paper 5). (8) The waves are anonymous (S_N gauge), and closure alone does not generate discrete species ($\mathcal{C}_k \simeq Q_{k-2}/S_k$, continuous for $k \geq 3$); what is discrete is the block size p that the floor quantizes, and the physical content resides in relational invariants (Paper 9).

These are organized along three unifying axes: **A. Unification of instability**—ignition qualification, SSB summit curvature, inflationary amplification, and Grassmann departure all begin from a single local instability $\rho(J_0) > 1$, and the tangent cocycle carries the subsequent real amplification. **B. Memory and erasure**—the transient records the initial structure in $J_0, \Phi(t, 0)$, but the terminal amplitude leaves only H/M and forgets the configuration, while the SSB history remains in the phase configuration and plane orientation (selective erasure of history information). **C. Relationalism**—neither an absolute wave number nor an absolute normal can be read; the physical content resides only in relational invariants modulo S_N , global phase, and coordinate rotation.

Note that the terms “inflation,” “vacuum,” and “spontaneous symmetry breaking” used in this paper are designations for mathematical features of this dynamical system; a substantive correspondence with cosmology or particle physics is open, as stated in §9.

This capstone stands on nine foundational papers (Papers 1–9). Its scope is limited to self-consistent single-coherent relative equilibria with a 90-degree skeleton and their small anisotropic perturbations; it does not enter into the identification of coordinates (whether $xyzt$ or including R, Q), the type of gauge (equal-interval / logarithmic / winding), the multi-particle case, or the understanding of the vacuum itself. These are explicitly noted as open (§9).

1. Introduction

This research series has numerically studied the dynamics of a closed system in which all pairwise relations $M = N(N - 1)/2$ among N entities are taken to be complex relational waves $z_e \in \mathbb{C}$, with external seeds removed. The waves live on the edges of the complete graph K_N , and the interaction is determined only by the phase differences between edges that share a vertex. The prior paper [1] reported this system's inflationary rapid expansion (exponential growth of the out-of-parent-plane component $H_\perp/H$, rank-4-like readout, transition to a metastable structure), but did not show **why a single relational wave takes that orbit** at the level of the equation of motion. Taking that single point as its starting point, this capstone establishes as a single causal chain the necessity of the initial value (the 90-degree floor), its qualification and exact solution, the reading of the rapid expansion as spontaneous symmetry breaking, the Grassmann geometry after locking, and the terminal amplitude universalization and relationalism.

The backbone of this capstone is **a single causal chain** running through Papers 1–9 (Papers 4, 8, and 9 are not auxiliary but the body of the chain).

- **(a) Dynamics** (§2, Paper 1): each edge wave couples in Kuramoto fashion with the vertex-sharing adjacent edges; the phase advances monotonically as $\sin^2$, and the amplitude increases or decreases via the second harmonic $\sin(2\Delta\varphi)$.
- **(b) Floor = 90-degree self-consistent relative equilibrium and its exact solution** (§3, §4, Papers 1 and 3): the maximally symmetric relative equilibrium that freezes the amplitude termwise is limited to the 90-degree lattice (the analytic reason why the termwise floor reached by `make_parent` becomes a 90-degree lattice; why the iteration selects that termwise branch is open, §4). On the 90-degree skeleton, $D^{-1}KD = iW$ makes the floor exactly equivalent to the Perron problem of the real symmetric W ; from the parity bipartite structure σ^2 , amplitude², and the full spectrum are closed form for **arbitrary** N , and the active dimension is $O(N)$.
- **(c) Ignition qualification and quantization of zero-closure** (§3.4, §3.5, Papers 2 and 4): the ignition qualification is “an unstable relative equilibrium” only (high symmetry, zero centroid, and square-closure are not qualifications). Zero-closure $\sum z^2 = 0$ alone is a continuous moduli, but the discrete symmetry of the cyclic-Fourier high-symmetry floor quantizes it into a block size $p = \text{spf}(N/\text{gcd}(N, 4))$ ($p = 2 \iff 8 \mid N$).
- **(d) Inflationary rapid expansion—from local instability to the tangent cocycle** (§5, §6, Papers 5 and 6): an infinitesimal seed breaks the symmetry, and the out-of-floor-plane component $H_\perp/H$ grows exponentially (**inflation**). The rapid expansion begins with the floor-Jacobian initial/frozen rate $\lambda_0 = \ln \rho_0$ ($\rho_0 = \rho(J_0)$, = the Landau summit curvature $V''(0) = -\lambda_0$), and its **real amplification rate is promoted to the finite-time Lyapunov rate γ_t of the tangent cocycle $\Phi(t, 0) = \prod_t J(z_t)$ along the real orbit** (for $N = 6, 8, 10$, $\gamma_t > \lambda_0$).
- **(e) Geometry—exponential departure on the Grassmann** (§7, Paper 7): the state is a rank-2 plane $P(t)$ at each instant ($\Pi_{t+1} = Q_t \Pi_t Q_t^T$), and since $H_\perp/H = \frac{1}{4} \|\Pi_t - \Pi_0\|_F^2$, inflation is exponential departure from the initial plane on $\text{Gr}(2, M)$; the terminal plane spans four dimensions with two nonzero principal angles (the cumulative map is not $SO(4)$).
- **(f) Terminal universalization and relationalism** (§8, Papers 5, 8, and 9): the terminus universalizes to equal amplitude $a_\infty = \sqrt{H/M}$ (an intensive terminal state quantity) and forgets

the configuration, while the SSB history remains in the phase configuration and plane orientation (selective erasure of memory). The waves are anonymous (S_N gauge), and the physical content resides only in relational invariants.

These (a)–(f) converge onto three unifying axes: **A. Unification of instability** (ignition qualification, SSB summit curvature, inflationary amplification, and Grassmann departure all begin from a single local instability $\rho(J_0) > 1$, and the tangent cocycle carries the real amplification), **B. Memory and erasure** (the transient records the initial structure, the terminal amplitude forgets, and the phase/geometry retains the history), **C. Relationalism** (neither an absolute wave number nor an absolute normal can be read; only relational invariants are physical).

What is introduced and used in this paper is only the dynamics defined in the prior paper (§2). Zero-closure and a particular amplitude distribution are not assumed but derived. Each section carries its claims and the key formulas/figures within this capstone, and delegates detailed derivations, exhaustive data, additional figures, and reproduction procedures to the corresponding foundational papers (Papers 1–9; Concept DOIs in Appendix A). The “density” of this capstone lies in integrating the nine papers as a single chain, not in duplicating raw data.

2. Dynamics and the single-wave equation of motion

2.1 The map

The `one_step` of the prior paper is given as follows. Let A be the adjacency matrix of the line graph $L(K_N)$ of K_N ($A_{ef} = 1$ if edges e, f share a vertex), $\varphi_e = \arg z_e$, then

$$K_{ef} = A_{ef} \sin(\varphi_f - \varphi_e)$$

is a real antisymmetric matrix (its entries are only the sines of phase differences, not containing the amplitude $|z_e|$). `one_step` is

$$z_{n+1} = \exp(\Delta\tau K) z_n, \quad \Delta\tau = \frac{2\pi}{\text{den}} = \frac{2\pi}{N}$$

and $\exp(\Delta\tau K)$ is a real orthogonal matrix. Because the same orthogonal matrix acts on the real and imaginary parts, the total norm $\|z\|^2$ and the square-closure $z^\top z = \sum_e z_e^2$ are exactly conserved.

Verification The error by which $\exp(\Delta\tau K(\varphi_n))z_n$ reproduces the canonical orbit z_{n+1} is at most 3×10^{-15} ($N = 6, 22, 40, \leq 300$ steps), confirming that K is precisely the generator (Paper 1).

2.2 The single-wave equation of motion

Writing the generator flow $dz/d\tau = Kz$ in components and decomposing $z_e = r_e e^{i\varphi_e}$, from the real and imaginary parts of $\sum_f A_{ef} \sin(\varphi_f - \varphi_e) r_f e^{i(\varphi_f - \varphi_e)}$, for adjacent edges $f \sim e$ we obtain

$$\boxed{\frac{dr_e}{d\tau} = \frac{1}{2} \sum_{f \sim e} r_f \sin(2(\varphi_f - \varphi_e))} \quad (\text{amplitude})$$

$$\boxed{r_e \frac{d\varphi_e}{d\tau} = \sum_{f \sim e} r_f \sin^2(\varphi_f - \varphi_e)} \quad (\text{phase})$$

Each edge wave is a Kuramoto-type oscillator that couples only with vertex-sharing adjacent edges via the sine of the phase difference. Since $\sin^2 \geq 0$, the phase advances monotonically (= rotates), and the difference between waves gives the phase displacement modulo rigid rotation. The increase/decrease of amplitude is driven by the second harmonic $\sin(2\Delta\varphi)$. In the frame modulo rigid rotation, the amplitude rearrangement appears as reorientation (precession) of the phase field, and this is the dynamical substance of the plane tilting during the inflation era (§7).

Verification The amplitude/phase components of the small substep $(\exp(\varepsilon K)z - z)/\varepsilon$ agree with the above equations to at most 10^{-6} (from rounding in the ε -scale difference) ($N = 6, 22, 40$, Paper 1).

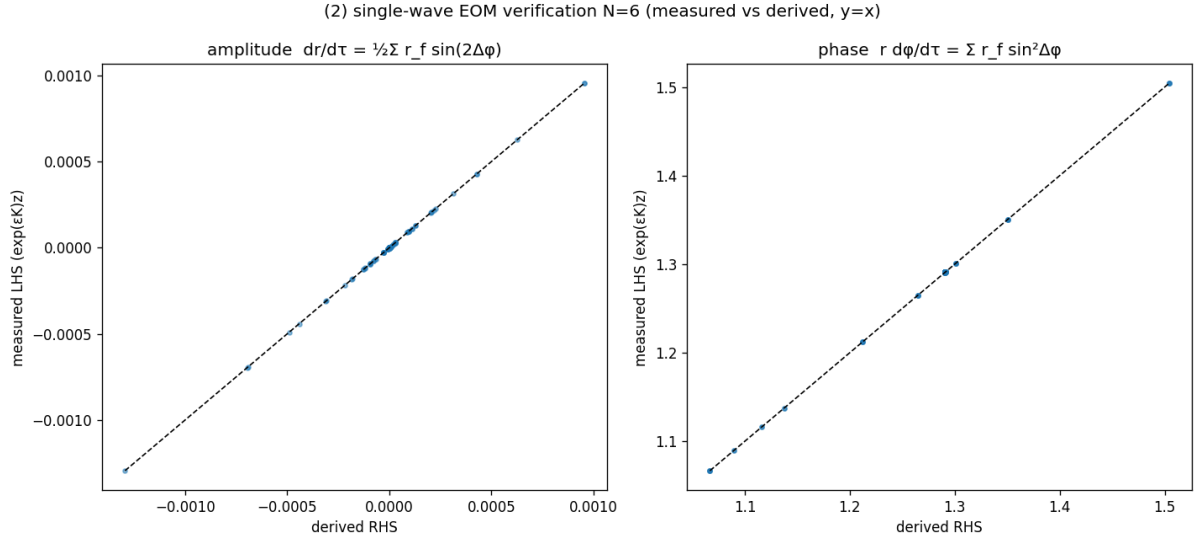


Figure 1: Verification of the single-wave equation of motion. The amplitude/phase components of the small substep $(\exp(\varepsilon K)z - z)/\varepsilon$ (measured LHS) agree with the derived equations (RHS) on $y=x$ ($N=6$).

Figure 1 Verification of the single-wave equation of motion ($N=6$, details in Paper 1). Left: amplitude, right: phase. The measurements (vertical axis) and the derived equations (horizontal axis) lie on $y = x$.

3. Derivation of the high-symmetry floor—the imposed conditions are the map plus the maximal-symmetry selection criterion

We show that the initial value (the floor) is not conveniently chosen for the result, as a derivation that merely imposes the selection criterion “relative equilibrium and maximally symmetric” on the map (there is no other input). This claim is doubly backed by the initial-value control experiments (Paper 2) and the algebraic exact solution (Paper 3).

3.1 The relative-equilibrium condition

The floor is a rigidly rotating relative equilibrium, i.e., it satisfies amplitude freezing $dr_e/d\tau = 0$ ($\forall e$) and phase velocity uniform over all e . From the amplitude equation, the former gives, as a necessary condition,

$$\sum_{f \sim e} r_f \sin(2(\varphi_f - \varphi_e)) = 0 \quad (\forall e) \quad (3.1)$$

(phase uniformity is automatically satisfied because, when the floor is taken to be an eigenmode of iK in §3.3, $z(t) = e^{-i\mu\tau} z_0$).

3.2 The 90-degree lattice is forced as the maximally symmetric relative equilibrium

There are two ways to realize (3.1): (i) each term vanishes individually (termwise), (ii) individual terms are nonzero but the sum cancels (cancellation). The **maximally symmetric equilibrium that annihilates all coupling terms individually** requires $\sin(2\Delta\varphi) = 0$, i.e., from

$$\sin(2\psi) = 0 \iff \psi = k \cdot 90^\circ \ (k \in \mathbb{Z})$$

that all adjacent phase differences are multiples of 90° , that is, all phases lie on a common 90-degree lattice $\{0, 90, 180, 270\}^\circ$ (modulo global phase), which is necessary and sufficient. This is not a choice but a consequence of the amplitude drive being the second harmonic (§4, Paper 1).

3.3 The amplitude is a deterministic eigenmode (algebraic exact solution)

Given the 90-degree lattice, the generator is fixed to an integer antisymmetric matrix (entries $0, \pm 1$), and the floor z is determined as its $\sigma_{\max}$ eigenmode (the eigenvector of the **most-negative eigenvalue in absolute value** $\mu = -\sigma_{\max}$ of iK ; the phase advances at $d\varphi/d\tau = -\mu = \sigma_{\max} > 0$, consistent with the phase equation of §2.2 $r\dot{\varphi} = \sum r_f \sin^2 \Delta\varphi \geq 0$; confirmed by real data and code). The amplitude is the square root of the eigenvalues of the integer symmetric matrix $K^\top K$ (= the singular values of the integer antisymmetric matrix K) = an algebraic number, a deterministic construction containing no continuous tuning. **More generally, for $D = \text{diag}(e^{i\varphi_e})$, on the 90-degree skeleton the identity $\sin \psi e^{i\psi} = i \sin^2 \psi$ gives $D^{-1}KD = iW$ ($W_{ef} = A_{ef} \sin^2 \psi_{ef}$, real symmetric, nonnegative, $0/1$), so the complex self-consistent floor problem $Kz = i\omega z$ ($z = Dr$) is exactly equivalent to the eigenvalue problem $Wr = \omega r$ of a real symmetric nonnegative integer matrix. If W is irreducible, by Perron–Frobenius the positive-amplitude principal eigenvector is unique ($\omega = \rho(W) = \sigma$)—that is, a self-consistent floor can be constructed on the 90-degree skeleton, and its uniqueness is guaranteed (Paper 1; the W of the 90-degree skeleton is the parity bipartite graph $W_{ef} = A_{ef} \mathbf{1}_{p_e \neq p_f}$, and Paper 3 proves that it is connected = irreducible for any $N \geq 3$). The integer-K floor is nothing but the closed form of the Perron eigenvector of this W . Zero-closure $\sum z^2 = 0$ also follows automatically on the 90-degree skeleton from $\|x\|^2 = \|y\|^2$ of this bipartite eigenmode (Paper 3, for all nonzero eigenmodes, not only the Perron one).** This construction has no free parameters, and σ^2 (even N : $N(N-2)$ / odd N : $N^2 - 2N - 1$) and each edge amplitude² (rational number, fixed by parity class) can be written in closed form (only the parity $p_e = (i+j) \bmod 2$ of the species $k_e = (i+j) \bmod 4$ makes $W_{ef} = A_{ef} \mathbf{1}_{p_e \neq p_f}$ bipartite, derived from the reduced Perron problem [even 2×2 / odd 3×3] for **arbitrary** N ; floor residual $\sim 10^{-15}$, sympy symbolic + numerical $N = 3\text{--}40$ verification, Paper 3). Furthermore, the active dimension is $O(N)$ (of the $M = O(N^2)$, the nonzero spectrum is $O(N)$ in count), and for $N \geq 6$ the closed form rank $K = 2(N-1)$, kernel dimension $(N-1)(N-4)/2$ holds ($N = 3, 4, 5$ are low-dimensional exceptions: $N = 3$ is rank 2 / kernel 1, $N = 4$ is rank 2 / kernel 4, $N = 5$ is rank 6 / kernel 4). The edge assignment of the amplitude class and the odd- N residual centroid, which were open in Paper 3, were also settled in closed form. The `make_parent` floor used by the experimental data (Papers 2 and 5–8) is an unstable self-consistent eigenmode on the same 90-degree phase lattice, but is a **different member of the relative-equilibrium moduli** with different amplitude distribution and centroid (centroid $|\sum z|$:

make_parent 0.25–1.41, integer-K floor ≤ 0.24). The integer-K floor is not the exact version of the data floor itself but an algebraically exact construction of a floor satisfying the same ignition qualification (differences and corrections in the section “Differences from and corrections to cited papers”).



図 2 Figure 2b: The 90-degree theoretical floor (integer-K $\sigma_{\max}$ analytic solution, all N=3..40). Phases 0/90/180/270, exact relative equilibrium (residual $\sim 1e-15$), 4|N gives exactly zero centroid.

Figure 2b Algebraic exact solution of the 90-degree floor ($N = 3-40$, details in Paper 3). All waves lie on two orthogonal axes, the 90-degree phase lattice.

3.4 Zero-closure is not a requirement but a consequence (Theorem)

Theorem If K is real antisymmetric and $v = a + ib$ is a self-consistent eigenmode $iK(v)v = \mu v$ ($\mu \neq 0$), then $\sum_e v_e^2 = 0$, that is, $a^\top a = b^\top b$ and $a^\top b = 0$.

Proof From the real and imaginary parts of $iK(a + ib) = iKa - Kb = \mu a + i\mu b$, $Ka = \mu b$, $Kb = -\mu a$. By antisymmetry, $\mu(a^\top a - b^\top b) = 0 \Rightarrow a^\top a = b^\top b$, and $\mu a^\top b = a^\top Ka = 0 \Rightarrow a^\top b = 0$. ■

The proof uses only “ K is real antisymmetric” and “ v is an eigenvector of iK ($\mu \neq 0$)”; self-consistency is the condition for v to be a floor and is not needed for the algebra. **Neither zero-closure nor any particular amplitude distribution is imposed on the construction of the floor. What is imposed is only the map and the selection criterion ‘maximally symmetric relative equilibrium (termwise $+\sigma_{\max}$)’, and the closure appears as a theorem.** (Measured: $\sum z^2 \sim 10^{-14}$ at the floor. Factorization into zero-closure blocks is §3.5.)

3.5 The factorization theorem of zero-closure and the 2-adic selection rule

On the cyclic-Fourier high-symmetry floor ($z_{ij} = r_d e^{i2\pi(i+j)/N}$), zero-closure $\sum z^2 = 0$ can be exactly factorized into smaller zero-closure blocks (this is a proposition about this high-symmetry floor and the root-of-unity structure of the same distance class, not about the data floor in general). The squared waves $w = z^2$ within a distance class cycle around a regular L -gon with phase step $q = e^{i8\pi/N}$ (order $L = N/\gcd(N, 4)$), and the minimal zero-closure block is the smallest prime factor of L ,

$$p = \text{spf}(L), \quad \boxed{p = 2 \iff L \text{ even} \iff 8 \mid N \iff \nu_2(N) \geq 3}$$

(minimality by the vanishing-sum theorem of Lam–Leung). For $N = 8k$ the minimal unit drops to two waves ($\pm i$ photon pairs), and otherwise it is a p -wave condensate (3 waves for $3 \mid N$, N waves for odd prime N). That $p = 2$ holds only for $8 \mid N$ is the **2-adic selection rule**: the square map doubles the phase step $4\pi/N \rightarrow 8\pi/N$, and the condition for antipodal points (the order-2 subgroup) to exist in $\mathbb{Z}_L$ is $L \text{ even} = \nu_2(N) \geq 3$ (Paper 4). Importantly, closure $\sum z^2 = 0$ **alone** has a continuous moduli as solution space ($\mathcal{C}_k \simeq Q_{k-2}/S_k$, $\dim_{\mathbb{C}} = k - 2$ for $k \geq 3$, Paper 9) and no discrete block size arises; it is quantized to $p = \text{spf}(L)$ **only when the discrete symmetry of the floor is imposed**. Note that the data floor `make_parent` has no exact $\pm i$ pairs (only near pairs, see the section “Differences from and corrections to cited papers”)——the factorization depends not on zero-closure or the 90-degree lattice but on the additional structure of the cyclic-Fourier floor.

3.6 No seed was planted——the ignition qualification is an unstable relative equilibrium

That the 90-degree floor is not a “special configuration that happens to work” is established by the initial-value control experiments (Paper 2). Replacing only the initial value with three families and running the same physics, (A) the symmetric parent with phases 0/90 (equal amplitude) and (B) the zero-centroid parent both do not ignite (a neutral fixed point, or immediate relaxation at step1); only (C) the high-symmetry theoretical floor (cyclic-Fourier relative equilibrium) and the experimental data’s `make_parent` floor ignite from the floor with an exponential ramp. That is, the **ignition qualification is “an unstable relative equilibrium (self-consistent eigenmode, residual at machine precision)”**

only; zero centroid, square-closure, and high symmetry themselves are not qualifications. The failure of (A)(B) is established to be “because it was **not an unstable relative equilibrium**,” not “because it was highly symmetric” ((A) also includes a neutral fixed point, and its not igniting demonstrates this distinction). This is the initial-value control version of the “amplification $\iff$ unstable relative equilibrium” of the onset-mode paper [2, Concept 21798854]. In addition, the seed that ignites the inflation is an infinitesimal, generic perturbation (whether floor residual $\sim 10^{-13}$ or random 10^{-8}), not a designed one (§5 exhaustive experiment). **This point is established by an experiment that directly verified the floor of the seed at 100-digit precision** (precision/seed-isolation experiment, $N = 7, 8, D = N$): even generating the initial value at 100 digits and lowering it to closure $\sim 10^{-102}$ and floor $f_{\perp} \sim 10^{-200}$, the amplification rate γ_{τ} **agrees to 8 digits ($N = 8$) / 6 digits ($N = 7$)** with the 64-bit run (= the exponential instability is intrinsic to the dynamics and depends on neither precision nor seed depth), it **necessarily ignites** after growing about 179 digits from the floor $\sim 10^{-200}$ by a single exponential law ($N = 8$ step 967 / $N = 7$ step 871), and the onset merely lags in proportion to the number of seed digits ($\Delta\text{step} = \Delta(\text{seed digits})/(\gamma_{\text{step}}/\ln 10)$). The float64 rounding during evolution is only a secondary effect advancing the onset by +6 steps. Therefore ignition is **neither a double-precision artifact nor a planted seed**, and the floor is a generically unstable starting point (the seed-independence of the amplification rate is also consistent with the prior inflation paper [Reference 1] and the seed-condition paper’s δ sweep 10^{-15} – 10^{-2} [Concept DOI 10.5281/zenodo.21874481]).

3.7 Established scope and open matters

The following are open because they have not yet been derived from this mathematical system (not out of timidity but because they are underived). What determines the overall scale r^2 of the floor is not within the framework (scale is privately held). The normalization $\|z\| = 1$ is a convention. Also, with a different generating measure a different self-consistent fixed point exists (the moduli of relative equilibria), and the 90-degree maximally symmetric floor is the most symmetric member of that family, not the unique solution. $N = 3$ splits in behavior by floor bifurcation: the T-type ($\sigma = \sqrt{2}$, $|\sum z| = 0.707$, ignites; make_parent floor / integer-K floor) and the triplet type ($\mu/r^2 = -3/2$, rank 1, neutral; cyclic-Fourier floor). The $N = 3$ that runs in the data series is the T-type and ignites (see the section “Differences from and corrections to cited papers”). These are explicitly noted as open.

4. Why the self-consistent floor becomes 90-degree—the reason make_parent naturally became 90-degree

We state the core of §3.2 independently. The amplitude drive is the second harmonic $\sin(2\Delta\varphi)$ arising from the $\sin(\varphi_f - \varphi_e)$ (first harmonic) of the generator, whose zeros on $[0, 360^\circ)$ are only $0, 90, 180, 270^\circ$ (confirmed by enumeration). Therefore the maximally symmetric self-consistent relative equilibrium that freezes the amplitude **termwise** (each term individually) is limited to phase differences of all pairs being multiples of 90° —a common 90-degree lattice. That make_parent converges for all N to a termwise 90-degree floor is an **empirical fact** (Experiment 4-1 / Study 28). The equation of motion in this section analytically explains **why the reached floor is always a 90-degree lattice: the maximally symmetric relative equilibrium that freezes the amplitude termwise is limited to the 90-degree lattice by $\sin 2\Delta\varphi = 0$ (Theorem)**. However, the basin / the iteration-selection mechanism itself, by which the make_parent iteration selects this termwise branch rather than the cancellation type,

is a separate question (open). Paper 1 treats the construction of the floor (self-consistent = eigenvalue problem of the amplitude) with the 90-degree as **given** (an empirical fact that naturally arose), and this capstone gives as a theorem the **reason** why the reached floor is a 90-degree lattice (termwise $\Rightarrow$ 90-degree)—the empirical given and the analytic necessity are two faces of the same fact.

Measured ($N = 6$) At the floor, the individual terms $|\sin(2\Delta\varphi)|$ of adjacent edges are at most 2.3×10^{-12} (termwise equilibrium). By contrast, at lock (the tilted vacuum), the individual terms are $O(1)$, at most 1.0, yet the weighted sum per edge is 1.4×10^{-7} (cancellation equilibrium), and the phase differences deviate from the 90-degree lattice by median 23.5° / max 44.6° . That is, the 90-degree lattice is “the unique maximally symmetric configuration where the amplitude drive vanishes for every term individually,” and the locked vacuum is a cancellation-type equilibrium distinct from it (Paper 1).

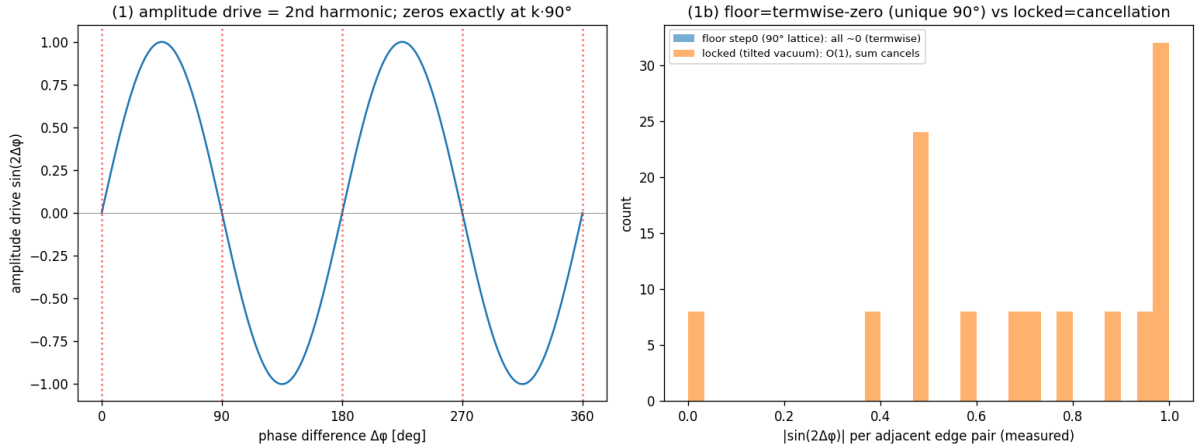


Figure 2: Left, the zeros of the amplitude drive $\sin(2\Delta\varphi)$ are limited to $k \cdot 90^\circ$. Right, the measured distribution of the adjacent-edge individual terms $|\sin(2\Delta\varphi)|$ for the floor (90-degree lattice, individual terms concentrated at 0 = termwise) and lock ($O(1)$, sum cancels = cancellation) ($N=6$).

Figure 2 90-degree necessity ($N=6$, details in Paper 1).

5. Spontaneous symmetry breaking and the degenerate vacuum

5.1 Mechanism

The floor that runs by SSB is a self-consistent unstable relative equilibrium with a 90-degree skeleton (the data floor make_parent, §3). An infinitesimal seed δ breaks the symmetry, is exponentially amplified along the unstable eigendirection of the floor (inflation, §6), and locks into a tilted relative equilibrium of equal amplitude and phase spread. This is a Mexican-hat spontaneous symmetry breaking: floor = an unstable maximum, valley = the family of degenerate relative equilibria (not a finite discrete symmetry orbit, §5.2), the seed spontaneously selecting one of its points. **What is called “vacuum” in this section is a mathematical designation for the degenerate minima of the effective potential of the order parameter r (= the locked relative equilibrium), not a characterization of the physical vacuum (§9).**

From the measured dynamics of the order parameter $r = \sqrt{H_\perp/H}$ ($H_\perp$ is the energy of the component

orthogonal to the floor plane), the Landau-type effective potential

$$V(r) = -\frac{1}{2}\lambda_0 r^2 + \frac{\lambda_0}{4r_{\text{vac}}^2} r^4$$

can be reconstructed with measured parameters ($\lambda_0 = \ln \rho_0$, $\rho_0 = \rho(J_0)$, $r_{\text{vac}} = \sqrt{H_{\perp}/H|_{\text{sat}}}$) ($N = 6$: $\lambda_0 = 0.338$, $r_{\text{vac}} = 0.485$). This is an effective description (ansatz) with coefficients identified from measured quantities. A first-principles derivation of $V(r)$ is left as future work (Paper 5).

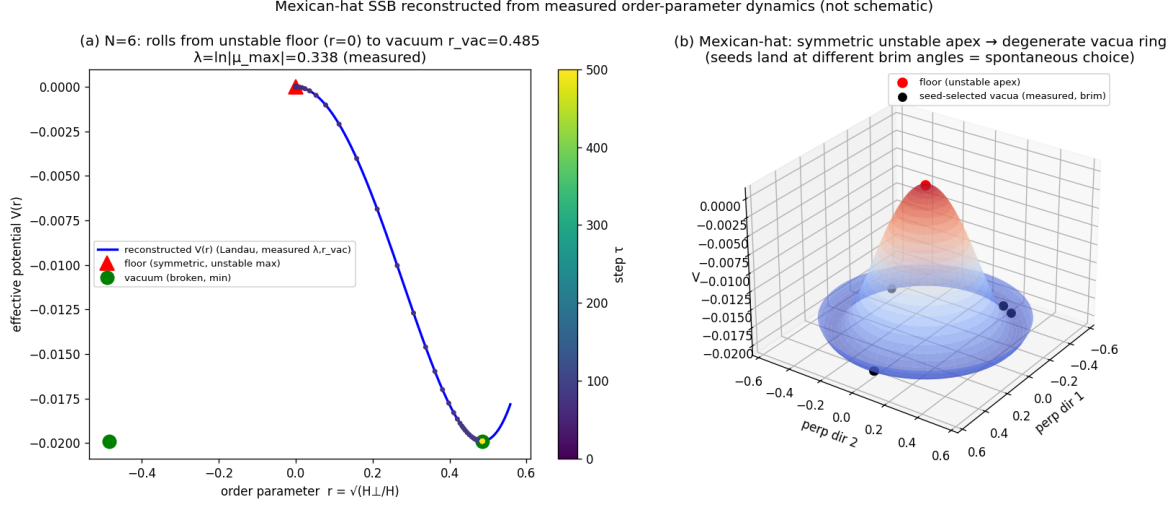


Figure 4: Figure 3a: $N=6$. On the reconstructed effective potential $V(r)$, the measured order parameter $r = \sqrt{H_{\perp}/H}$ falls from the floor ($r=0$, unstable summit) to the valley $r_{\text{vac}}=0.485$ (color = step). On the sombrero surface, an unstable summit and a degenerate vacuum ring (5 seed points).

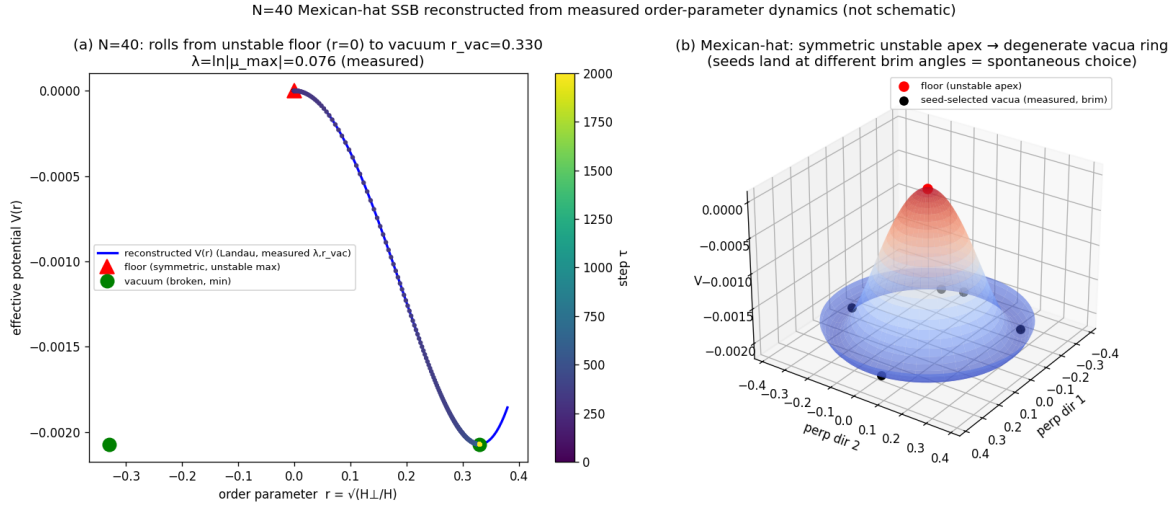


Figure 5: Figure 3b: $N=40$. Same procedure. $\lambda_0=0.076$, $r_{\text{vac}}=0.330$. Fall from floor to valley + degenerate vacuum ring.

Figure 3 Mexican-hat SSB (measured reconstruction, details in Paper 5). Top = $N=6$ ($\lambda_0 = 0.338$, $r_{\text{vac}} = 0.485$), bottom = $N=40$ ($\lambda_0 = 0.076$, $r_{\text{vac}} = 0.330$).

5.2 Exhaustive verification ($N = 6, \dots, 40$)

To the same floor Z_0 , five infinitesimal seeds of amplitude 10^{-8} in different orientations were applied, and each was run to adaptive lock by the exact one_step. In all $35 \text{ values} \times 5 \text{ seeds each} = \mathbf{175 \text{ runs}}$ for $N = 6, \dots, 40$: - Every seed causes inflation (clear departure from the floor). - All lock into an isomorphic equilibrium with equal amplitude ($|a_{\max}/a_{\min} - 1| \lesssim 10^{-3}$) and a 1-step rotation angle matching $\Delta_{\text{tail}} = -2\pi/N$ (error $\sim 10^{-5}$). - The phase configuration (vacuum orientation) differs by seed (median vacuum difference $4^\circ\text{--}48^\circ$).

That is, from the unstable floor, an infinitesimal seed spontaneously selects one of the degenerate vacua. This is the spontaneous symmetry breaking of this system (**confirmed in 175/175**, Paper 5). The **degenerate vacuum family is not identifiable at least as a finite discrete orbit consisting of S_N + global phase + complex conjugation**: it was confirmed that the five terminal states for $N = 6$ do not map to one another by any of the $6! = 720$ vertex permutations + global phase + complex conjugation (minimal residual 0.05–0.83, Paper 5). **The point of this result is that from the same 90-degree floor the terminal plane (vacuum orientation) is not uniquely determined but is seed-dependent.** The existence/identification of a larger continuous symmetry group, and the continuity of the degenerate vacuum family (whether a 1-parameter continuous scan of the seed direction $\delta(\theta) = \cos \theta \delta_1 + \sin \theta \delta_2$ moves the terminus continuously), are open (this open matter does not demote SSB itself).

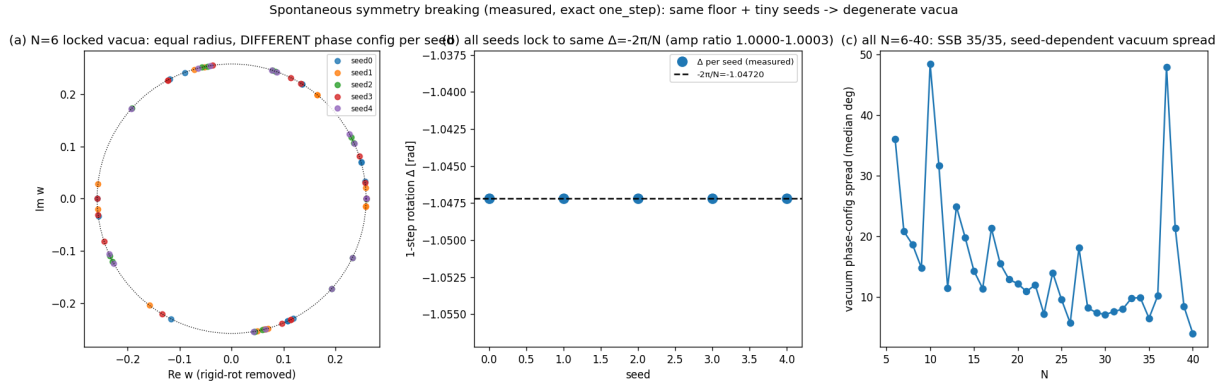


Figure 4: $N=6$, same floor + five different infinitesimal seeds. Left = the locked configuration of each seed (rigid rotation removed; equal radius but differing phase configuration = degenerate vacuum), center = all seeds match $\Delta = -2\pi/N$, right = the N -dependence of the vacuum difference over all $N=6\text{--}40$.

Figure 4 SSB degenerate vacua (real data, $N=6$).

Figure 5 SSB over all $N = 6\text{--}40$ (175/175 confirmed).

Figure 5a Inflation $H_{\perp}/H$ for all $N = 3\text{--}40$ (step500, same arrangement as Figure 5b). In each panel the $\text{den} = N$ ($2\pi/N$) curve is this system's inflation, ramping exponentially from the floor ($\sim 10^{-29}$, float64 rounding is the seed) $\rightarrow$ saturation. As N increases, the rate drops and the onset lengthens (§6). The other curves are denominator-control controls ($2\pi/(N \pm k)$, $2\pi/124$). From Figure 5a (inflation) $\rightarrow$ Figure 5b (terminal complex), floor $\rightarrow$ rapid expansion $\rightarrow$ lock can be surveyed for all N at a glance.

Figure 5b Terminal complex diagram (make_parent data floor, step500, all $N = 3\text{--}40$). The floor

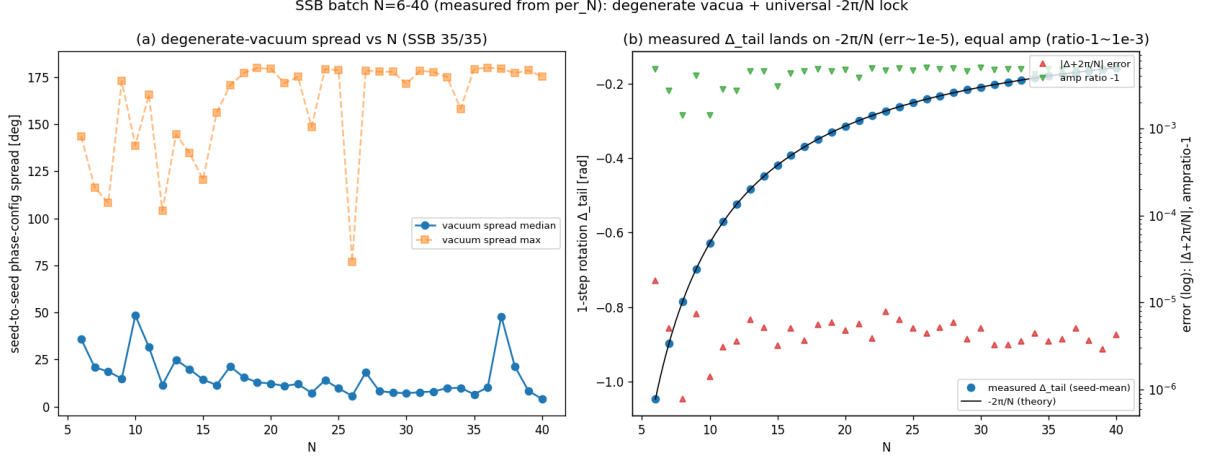


Figure 5: The SSB batch over all $N=6-40$. Left = the N -dependence of the vacuum difference (median/max), right = the measured Δ_{tail} matches the theoretical $-2\pi/N$ for all N (error $\sim 1e-5$), amplitude ratio $\sim 1e-3$.

(Figure 2b of §3.3, and `make_parent step0` = Figure 1 of Paper 2) has locked by SSB into a tilted relative equilibrium of **equal amplitude** (radius $\sim 1/\sqrt{M}$) and $\Delta = -2\pi/N$. Against the petal-like `step0` of the 90-degree lattice, the terminus has each wave spread on a ring of common radius (the $a_\infty = \sqrt{H/M}$ of §8). Small N is already locked, and large N (≥ 18) is still filling the ring at `step500` and locks entirely by 2000 steps (§5.3). The integer- K analytic floor (Figure 2b) also locks into the same universal terminus ($\Delta = -2\pi/N$, equal amplitude).

5.3 Long-time verification ($N = 22, 40, 2000$ steps)

The large N ($N = 22, 30, \dots, 40$) that was unlocked at `step500` was due to truncation, and running without stopping to 2000 steps locks them all. $N = 22$ fully locks at `step 663`, and $N = 40$ at `step 1299` (Δ_{tail} exactly $-2\pi/N$, equal-amplitude ratio 1.0000, floor/tail rigid-rotation residual machine-zero). A control test confirmed that the initial value bit-matches the old run at `step0` (`step1` difference $\sim 10^{-16}$, thereafter inflation exponentially amplifies), ensuring faithful reproduction (Paper 5).

The inflation curve (exponential growth of $H_\perp/H \rightarrow$ saturation) and the terminal `step2000` complex diagram are shown for $N = 22, 40$.

Figure 5c Inflation $H_\perp/H$ ($N = 22, 2000$ steps). Grows by exponential ramp from the floor ($\sim 10^{-29}$, float64 rounding is the seed) and saturates.

Figure 5d Inflation $H_\perp/H$ ($N = 40, 2000$ steps). Exponential ramp $\rightarrow$ saturation (as N increases the rate drops and the onset lengthens, §6).

Figure 5e Terminal complex diagram ($N = 22, \text{step2000}$). All $M = 231$ waves are spread on a ring of common radius $\sim 1/\sqrt{231} =$ fully locked (`step663`).

Figure 5f Terminal complex diagram ($N = 40, \text{step2000}$). All $M = 780$ waves are spread on a ring of common radius $\sim 1/\sqrt{780} = 0.0358 =$ fully locked (`step1299`, $\Delta_{\text{tail}} = -2\pi/40$). Figure 5b (large N at `step500` still filling the ring) locks all around by 2000 steps as in Figures 5e and 5f.

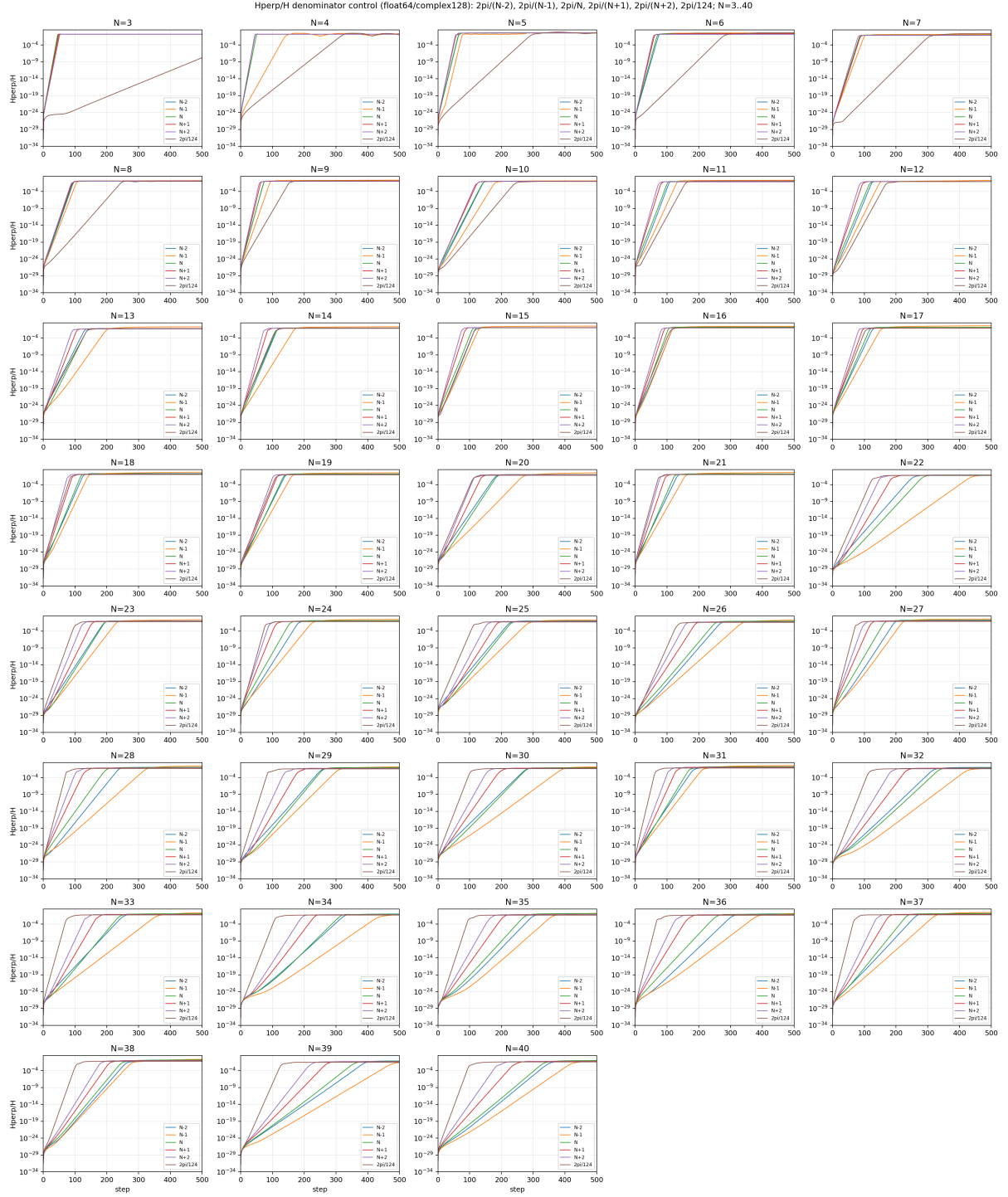


Figure 5a: The inflation $H_{\perp}/H$ (step500) for all $N=3-40$ in the same arrangement as Figure 5b. The $\text{den}=N$ ($2\pi/N$, canonical) curve is this system's inflation, ramping exponentially from floor $\sim 1e-29 \rightarrow$ saturation. The others are denominator-control controls.

make_parent parents sweep: complex plane at final step 500 (den=N); N=3..40

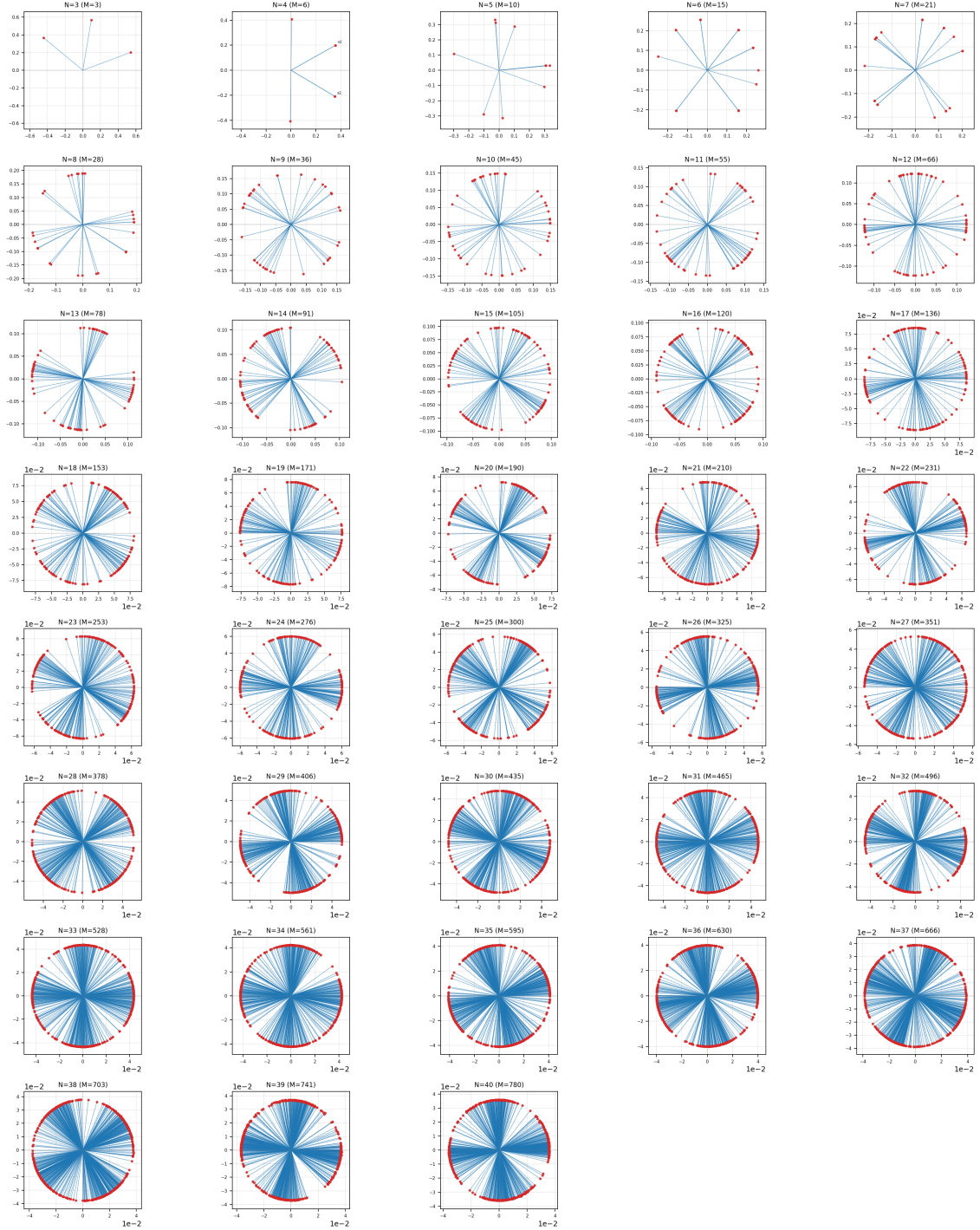


Figure 5b: The terminus (step500) complex plane of the make_parent data floor (all $N=3..40$). Equal-amplitude ring (radius $\sim 1/\sqrt{M}$) with the phase spread = a locked tilted relative equilibrium. Small N is already locked, large N is still filling the ring (all lock by 2000 steps, §5.3).

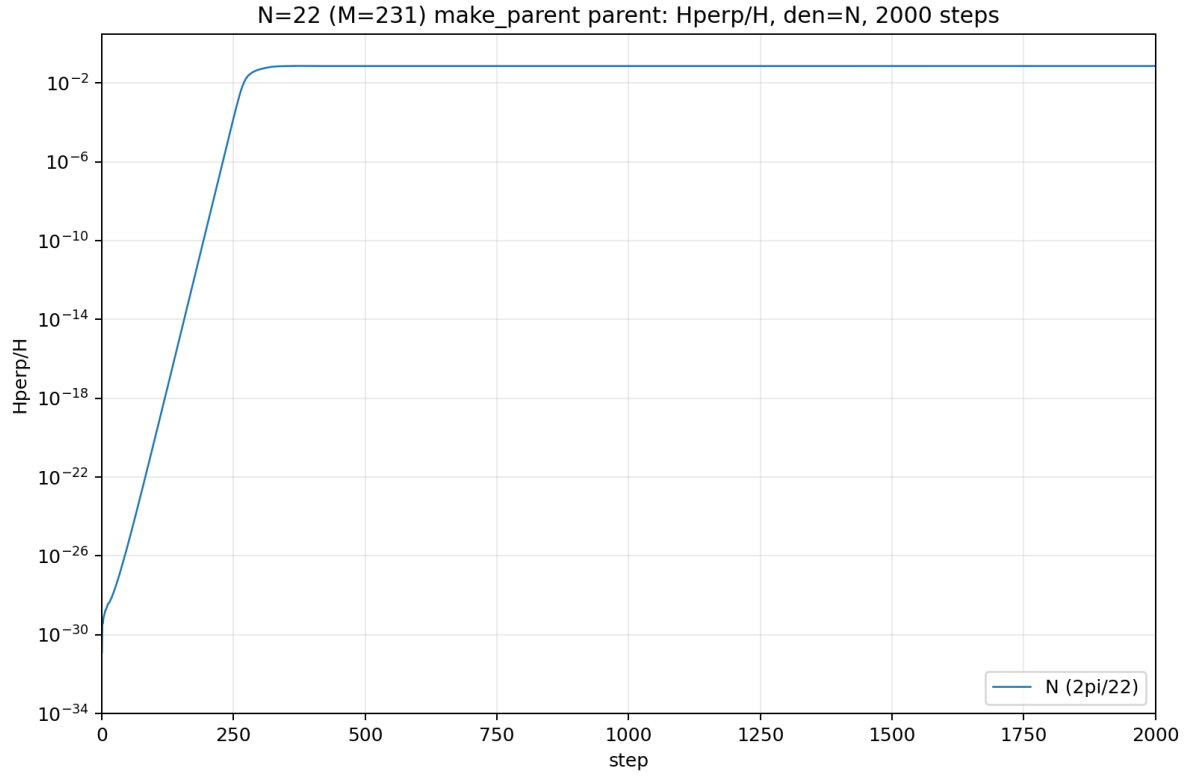


Figure 5c: $N=22$ inflation $H_{\perp}/H$ (2000 steps). Grows by exponential ramp from the floor ($\sim 1e-29$) and saturates.

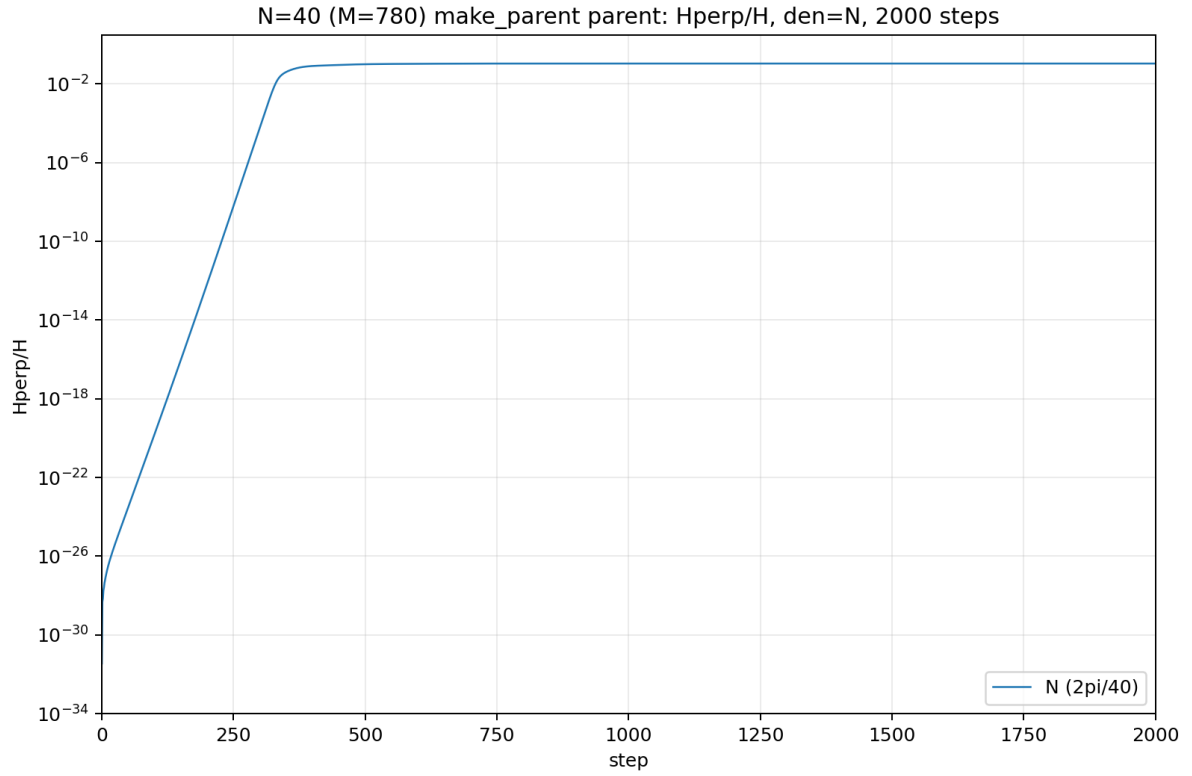


Figure 5d: $N=40$ inflation $H_{\perp}/H$ (2000 steps). Similarly exponential ramp $\rightarrow$ saturation (saturation $\sim 3e-2$).

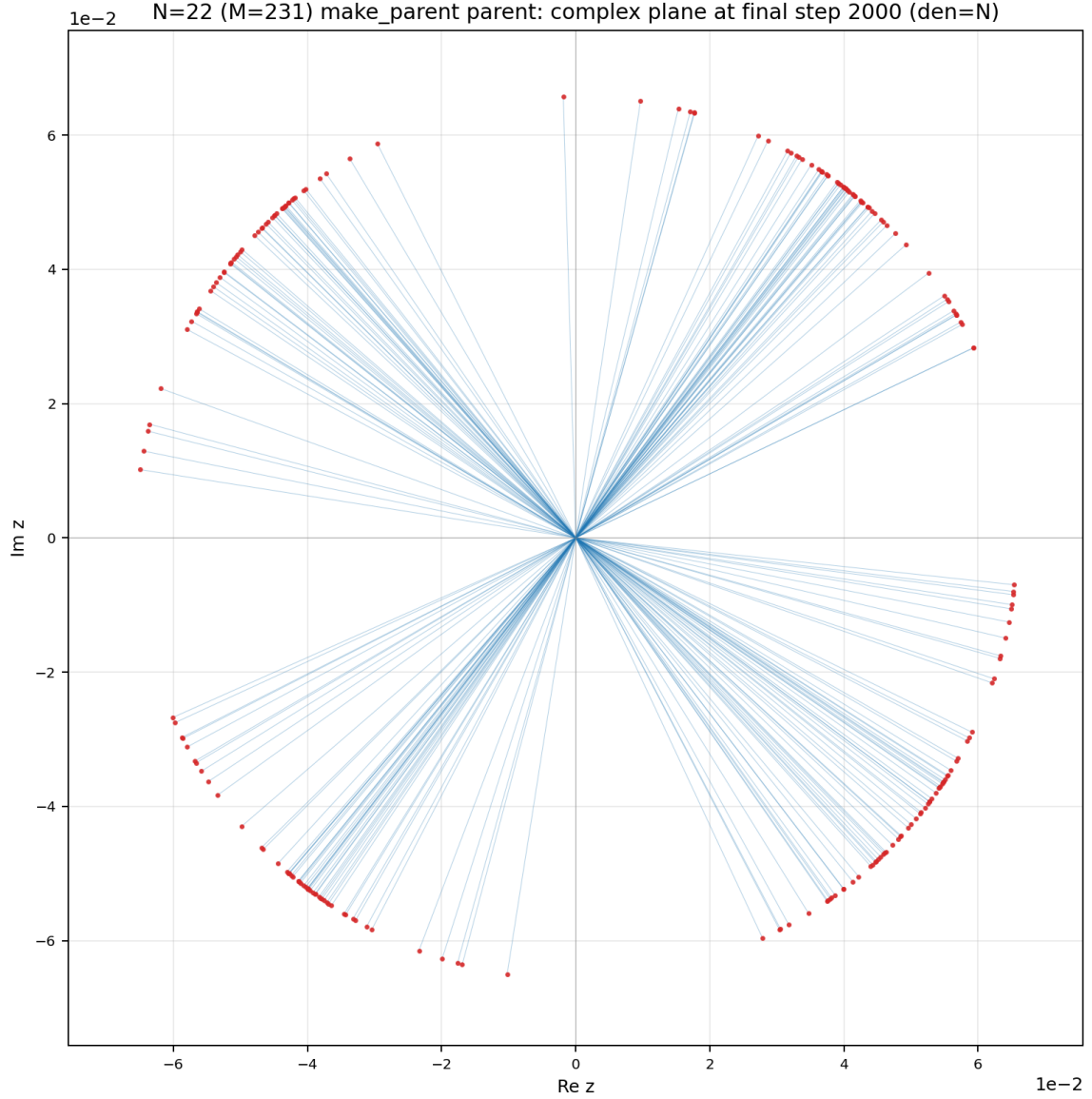


Figure 5e: $N=22$ terminal step 2000 complex plane. Equal-amplitude ring (radius $\sim 1/\sqrt{231}$) with the phase spread = fully locked.

6. The inflation curve = linear instability of the floor

The inflation curve is the linear-instability amplification of an infinitesimal seed. Write the spectral radius of the floor's linearized map (Jacobian J) as $\rho_0 = \rho(J_0)$, $\lambda_0 = \ln \rho_0$ (a quantity distinct from the iK eigenvalue μ of §3; henceforth the Jacobian side is consistently ρ_0, λ_0). ρ_0 gives

$$\frac{H_{\perp}}{H}(t) \approx H_{\perp 0} \rho_0^{2t} \quad (H_{\perp 0} = \text{seed}^2), \quad \text{ramp slope} = 2 \log_{10} \rho_0, \quad \text{onset} \approx \frac{\ln(1/H_{\perp 0})}{2 \ln \rho_0} = \frac{\ln(1/\text{seed})}{\lambda_0}$$

The slope of $H_{\perp}$ obtained by evolving a small δ under the fixed floor J matches $2 \log_{10} \rho_0$ exactly ($N = 6$: 0.2935 vs 0.2937). ρ_0 decreases with N ($N = 6$: 1.402, 8: 1.269, 10: 1.185), so inflation becomes slower and the onset lengthens (measured onsets are 64, 87, 133 steps for $N = 6, 8, 10$). A closed-form N -dependence

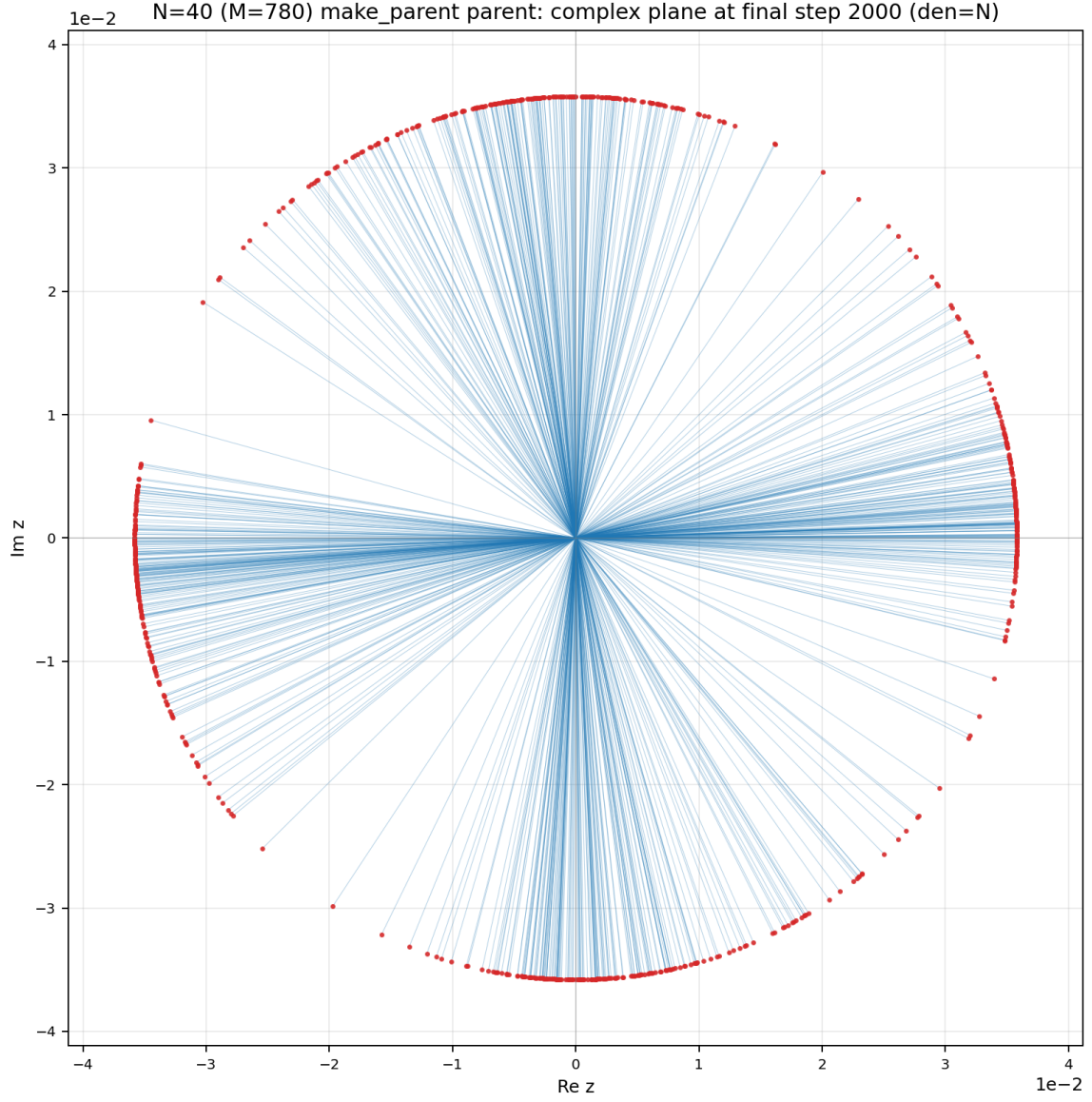


Figure 5f: N=40 terminal step2000 complex plane. All 780 waves on a ring of radius $\sim 1/\sqrt{780}$ = fully locked.

law of the onset is left as future work.

The amplification is a **two-stage dynamics**. Near the floor $\delta_{t+1} = J_0 \delta_t$ with the initial/frozen rate $\lambda_0 = \ln \rho_0$ (= the Landau summit curvature $V''(0) = -\lambda_0$, §5). Away from the floor the state itself changes and $\delta_t = \Phi(t, 0) \delta_0$, $\Phi(t, 0) = J(z_{t-1}) \cdots J(z_0)$ (the **tangent cocycle** = the time-ordered product of the time-varying, noncommuting Jacobians). **The real inflation rate is the finite-time Lyapunov rate of this tangent cocycle**

$$\gamma_t = \frac{1}{t} \ln \frac{\|P_{\perp} \Phi(t, 0) \delta_0\|}{\|P_{\perp} \delta_0\|}, \quad \text{ramp slope} = \frac{2\gamma_t}{\ln 10}$$

($P_{\perp}$ = the orthogonal projection onto the floor plane). Reproduction A **directly confirmed** that the

$H_\perp$ slope evolved by $\prod_t J(z_t)$ along the real orbit matches the measured ramp (time-varying / measured = 0.999/1.004/1.019, $N = 6, 8, 10$). J is non-normal (0.53–0.76) and noncommuting, so $\Phi(t, 0) \neq J_0^t$ and the direction of maximal amplification is amplified while rotating. $\lambda_0 = \ln \rho_0$ (the initial/frozen rate near the floor) and the real rate γ_t are distinct quantities, and what carries γ_t is $\Phi(t, 0)$ (the amplification dynamics of the real orbit). For $N = 6, 8, 10$, $\gamma_t > \lambda_0$ (the real ramp exceeds the floor value by 18–31%)—not a “correction” but a consequence of the two-stage dynamics. That γ_t does not in general fall below λ_0 (that λ_0 is a lower bound) is limited in this system to the observations for $N = 6, 8, 10$ and has no general proof. A closed form of γ_t is also underived (Paper 6).

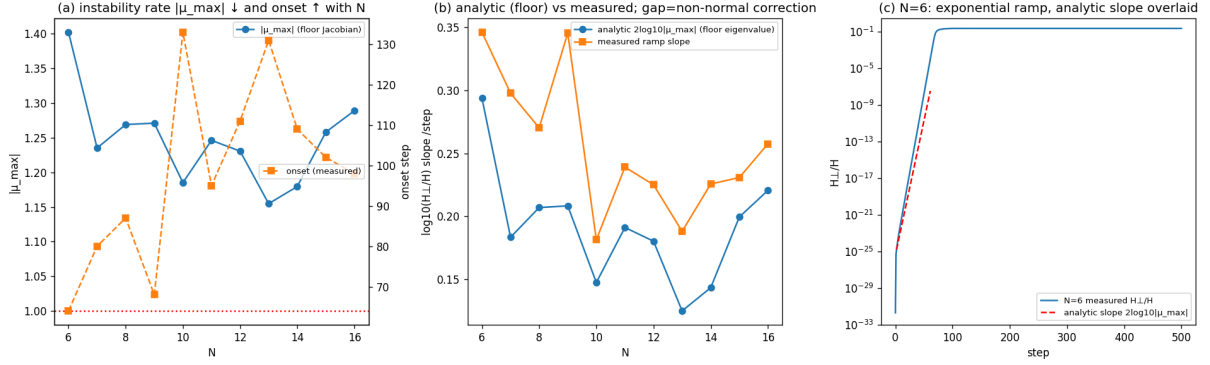


Figure 6: Inflation = linear instability. Left = the N -dependence of the floor-Jacobian $\rho_0 = \rho(J_0)$ and the onset (larger N decelerates, onset increases), center = the analytic ramp slope $2\log_{10} \rho_0$ (floor) vs measured (non-normal gap), right = the analytic slope superimposed on the measured $H_\perp/H$ ramp for $N=6$.

Figure 6 Inflation = linear instability of the floor (details in Paper 6).

7. Geometry after locking—plane switching on the Grassmann and the 4-dimensional span of the initial and terminal planes

We establish what inflation and locking do in state space—the **instantaneous rank-2 plane** $P(t)$ **departs exponentially from the floor plane on the Grassmann manifold** $\text{Gr}(2, M)$ **and switches to a different rank-2 plane**. This is the geometric consequence of the dynamics of §2 (amplitude rearrangement = reorientation of the phase field = precession).

(i) **Instantaneous rank 2 and Grassmann motion** The map is a real orthogonal rotation $z_t = M_t z_0$ ($Q_t = \exp(\Delta\tau K_t)$ is the one-step map, $M_t = Q_{t-1} \cdots Q_0$ the cumulative map, both real orthogonal acting identically on real and imaginary parts), and the state stays on a 2-dimensional plane (rank 2) at every instant (the singular values of $[\Re z, \Im z]$ are only two-valued at all steps; by conservation of $z^\top z = 0$ an orthogonal isometric frame). Modulo global phase, the observable is the 2-dimensional subspace $P(t) = \text{span}\{\Re z, \Im z\}$ itself (motion on $\text{Gr}(2, M)$), and the projection Π_t moves exactly under the one-step map Q_t as $\Pi_{t+1} = Q_t \Pi_t Q_t^\top$. At the floor $Q_0 P_0 = P_0$ ($Ka = \mu b$, $Kb = -\mu a$) is fixed (principal angles zero), during inflation $\Pi_t \neq \Pi_0$ moves, and at saturation it re-locks to P_{term} .

(ii) **The initial and terminal planes span four dimensions** The locked plane P_{term} tilts from P_0 , and the principal angles are, independently of N , **both nonzero** ($N = 6$: $27.1^\circ, 30.8^\circ$, $N = 40$: $18.1^\circ, 20.4^\circ$), with $\dim(P_0 + P_{\text{term}}) = 4$. The tilt of the locked plane from the floor is 14.2° – 29.0° (median $\sim 19.2^\circ$) over

all $N = 6\text{--}40$. **From the same floor this terminal orientation is not uniquely determined but is seed-dependent (§5).**

(iii) The tilt, the two principal angles, and $H_\perp/H$ are the same Grassmann quantity (an identity holding at all times) By definition, for the principal angles $\theta_1(t), \theta_2(t)$ between P_0 and $P(t)$ at each time t ,

$$\frac{H_\perp(t)}{H} = \frac{1}{2} (\sin^2 \theta_1(t) + \sin^2 \theta_2(t)) = \frac{1}{4} \|\Pi_t - \Pi_0\|_F^2$$

($\Pi = UU^T$) holds (not only for the terminal Π_{term} but for all t). Hence from the initial ramp $H_\perp/H \propto e^{2\gamma_t t}$ of §6, immediately

$$\|\Pi_t - \Pi_0\|_F \propto e^{\gamma_t t}$$

—**inflation is exponential departure from the floor plane on $\text{Gr}(2, M)$.** This is not “it can be read that way” but a **derived** geometric description combining the identity from the definition with the exponential rate of §6.

(iv) The 4D span is a minimal span, not $SO(4)$ dynamics During the transition the second direction rises (transient $s_3/s_1 \approx 0.10\text{--}0.20$ = the local span swept in a short time by the two moving planes), and after lock it returns to a single plane (tail $s_3/s_1 \approx 0$). The cumulative map $M_t = \prod \exp(\Delta\tau K)$ does not preserve the 4-dimensional subspace ($M_t S \neq S$, confirmed by a discriminating computation), so **“the dynamics is an $SO(4)$ double rotation on four dimensions” is rejected**— $\dim(P_0 + P_{\text{term}}) = 4$ is the minimal span needed for the relative configuration of two 2-planes, not an invariant space in which the dynamics lives. A short-window SVD test separated out that the plane tilt is real, not an artifact of long-window aggregation (Paper 7).

Note that when initially the complex plane is only a single sheet, its normal is in a degenerate orthogonal complement and is unobservable (conceptually). Only when inflation raises a specific orientation in the normal space does the relative phase (tilt) between the original plane and the new plane become readable. What can be read is always a relational quantity (the relative phase between planes); the absolute normal direction cannot be recovered from the state alone (Q_t hides the orientation, and no conserved quantity points to it).

8. Terminal universality, selective erasure of memory, and anonymity

Terminal universality (intensive state quantity): after locking, the terminus is of equal amplitude, and its radius is determined only by the extensive variables (H, M) : $a_\infty = \sqrt{H/M}$. This is an exact consequence of $H = \sum_e |z_e|^2 = M a_\infty^2$; (H, M) are extensive quantities, $\rho = H/M$ is intensive, and $a_\infty = \sqrt{\rho}$ is an **intensive terminal state quantity**. What is nontrivial is not the formula but the **terminal universality**—a mixed system with the triangular-number mixture $M(9) = M(7) + M(6)$ normalized to the same density, and a single system, both converge to the same $a_\infty = 1/6$; and even though the presence/absence of ignition itself changes by initial-value family (staged / analytic floor / v2) (v2 is a non-igniting neutral fixed point), the terminus is common (Paper 8).

Selective erasure of memory (Axis B): the transient (inflation era) records the initial structure in the floor Jacobian J_0 and the tangent cocycle $\Phi(t, 0) = \prod_t J(z_t)$ and determines the onset and path. But after nonlinear saturation $|z_e|^2 \rightarrow H/M$, and **the configuration information disappears from the amplitude sector**. On the other hand, the SSB history remains in the phase configuration and the terminal plane orientation (§5, §7). Therefore

the amplitude universalizes to H/M and forgets the configuration, while the phase configuration and plane orientation

——not “forgetting everything” but a **sector-selective erasure** (selective erasure of history information).

Anonymity and relationalism (Axis C): the edge numbering is a gauge of the vertex permutation S_N , and only S_N -invariants can be read (“which wave” cannot be read). At the terminus even the individuality of $|z_e|$ disappears, so they cannot be distinguished even by value, and the indistinguishability of identical particles appears as a consequence of the dynamics. This is the same principle as the relational readout of §7 (the absolute normal is unobservable, only the relative phase between planes can be read), and **the physical content resides only in relational invariants modulo S_N , global phase, and coordinate rotation** (Paper 9).

Closure does not generate discrete species——the quantization is carried by the floor: even under the **hypothesis** that the minimal closure block, not the individual wave, is the unit of information, closure $\sum z^2 = 0$ **alone** does not generate discrete species (a mathematical fact): $\mathcal{C}_k \simeq Q_{k-2}/S_k$ (the projective quadric hypersurface $\dim_{\mathbb{C}} = k-2$), where $k=2$ has $[1:i], [1:-i]$ collapsing into one orbit under S_2 ($\pm$ is not an anonymous bit), and $k \geq 3$ is continuous. Q_{k-2} is connected for $k \geq 3$, and the quotient by the finite group S_k is also connected, so from closure alone **no “species” as a discrete connected component arises**. What is discrete is the block size $p = \text{spf}(L)$ of §3.5——**closure + the discrete symmetry of the cyclic-Fourier high-symmetry floor quantizes the block size** (Paper 9). The identification of the self-consistent dynamics (orientation, winding, topology, dynamical selection rule) that selects a discrete type from the continuous relational moduli is the next research problem.

9. Established scope and open problems

The following are open because they have not yet been derived from this mathematical system (not out of timidity but because they are underived).

- **Identification of coordinates:** whether the 4-dimensional span obtained in §7 can be identified with physical $xyzt$ (or $xyztRQ$, $xyztRQ_1Q_2Q_3$) is open. What is established is that the dimension of the minimal span spanned by the initial 2 planes and the terminal 2 planes is $\dim(P_0 + P_{\text{term}}) = 4$, and that their relative configuration has two nonzero principal angles $\theta_1, \theta_2 \neq 0$ ——not four physical axes, and not two independent rotational dynamics ($SO(4)$ dynamics is rejected in §7).
- **Gauge (graduation):** whether a gauge can be defined along each direction, and its type (equal-interval / logarithmic / winding), is undefined. The external step τ is not a physical internal clock (prior work measured that the equal-spacing of τ breaks after inflation), and whether an intrinsic equal-interval clock exists is open. A clock stands only as the beat of a coherent, distinct-frequency two-component; it does not stand for a single coherent mode.

- **Narrowness of the state’s symmetry:** this research primarily targets a self-consistent single-coherent relative equilibrium with a 90-degree skeleton and its small anisotropic perturbations. States with coexisting particles of various masses and charges are not treated.
- **Vacuum:** whether the 90-degree floor is the true vacuum or a condensate that already has structure is unspecified. The vacuum itself is not characterized.
- **Caveat on circular derivation:** the terminal equal amplitude (Paper 8) inherently involves this engine’s phase normalization (the processing that kills the amplitude), so it is not presented as an independent corroboration of the norm form. Independent verification in an amplitude-preserving control dynamics is open.
- **The band/hair register of Paper A is scaffolding (demotion of the assessment):** the types (boson/fermion, etc.) read in the prior work (Paper A, Concept 21874481), which gives each wave $16 \times 8 = 128$ internal registers, are scaffolding by **declared constants** not derived from the closure axiom or anonymity, and this series does not cite them as a physical claim. To physicalize the types, one must derive a discrete degree of freedom from the closure axiom + the discrete symmetry of the floor + a dynamical selection rule (currently not achieved, Paper 9).
- **Quantization mechanism of discrete species:** the self-consistent dynamics (orientation, winding, topology, dynamical selection rule) that selects a discrete type corresponding to a particle species from the continuous closure moduli ($\mathcal{C}_k \simeq Q_{k-2}/S_k$) is unidentified (§8, Paper 9).
- **Others:** the privately-held scale r^2 , the moduli of relative equilibria, the $N = 3$ exception, and the non-readability of the absolute time t .

10. Related work (structural agreement——correspondence to prior art, not the derivation source)

The results of this research were derived independently of the map of §2, but their structure agrees with existing theory.

- **Kuramoto model** (Kuramoto 1975): the generator $K_{ef} = A_{ef} \sin(\varphi_f - \varphi_e)$ is a Kuramoto-type phase coupling on $L(K_N)$, and locking corresponds to synchronization.
- **Spontaneous symmetry breaking / Landau theory / Goldstone-Higgs** (Landau 1937; Goldstone 1961; Anderson 1963; Higgs 1964): the phenomenon of §5 is an instance of SSB on a discrete map, and the reconstructed effective potential is a Ginzburg–Landau-type sombrero.
- **Transient amplification of non-normal matrices / pseudospectra** (Trefethen & Embree 2005): structural agreement of the transient amplification of a fixed non-normal system. In this system, the amount by which the real ramp persistently exceeds the floor eigenvalue is carried not by the transient of the fixed floor but by the time-varying, noncommuting tangent cocycle $\Phi(t, 0)$ along the real orbit, as confirmed in Reproduction A (§6).
- **Spectra of line graphs** (Cvetković et al., algebraic graph theory): the eigenvalues of $A = L(K_N)$.
- **Relative equilibria** (Marsden & Ratiu, geometric mechanics): the rigidly rotating relative equilibria of the floor and lock.

Differences from and corrections to cited papers (notes for consistency)

There are three discrepancies between this capstone and the foundational papers regarding the identification of the floor. So that the claims of the capstone do not waver, we explicitly state the claim on the original-paper side, and that it is erroneous (or how it should be read with restriction), with the reason from derivation/measurement. The description of the original papers at the time of writing is left without weakening, and errors are corrected here as errors. What is established is that the ignition qualification is “an unstable relative equilibrium” only, and that the floors satisfying it split into three mutually distinct families: (i) the **make_parent floor** used by the experimental data (Papers 5–8) (90-degree phase skeleton, unstable relative equilibrium, nonzero centroid, no exact $\pm i$ pairs), (ii) the **integer-K analytic floor** of Paper 3 (an exact relative equilibrium on the same 90-degree phase skeleton, centroid ≤ 0.24 , with $\pm i$ pairs for $N = 8k$), (iii) the **cyclic-Fourier floor** of Paper 2(C) / Paper 4 (a different high-symmetry relative-equilibrium family, generally not a 90-degree phase lattice but based on cyclic symmetry— $z_{ij} = r_d e^{i2\pi(i+j)/N}$). (i)(ii) are distinct members sharing the 90-degree skeleton, and (iii) is a third family whose skeleton itself differs.

Correction 1: the implication of Paper 3 that “the integer-K floor is the exact version of the make_parent floor (the same floor)” turned out to be erroneous. Paper 3, in its Abstract, §1, and §4 (precision comparison table), treated the make_parent floor as “a solution approximating the self-consistent eigenmode by iteration (residual $\sim 10^{-13}$),” and, placing the integer-K floor’s relative-equilibrium residual $\sim 10^{-15}$ as “exceeding the $\sim 10^{-13}$ of make_parent,” presented the integer-K floor as **a more precise construction of the same floor**. This implication of “the same floor” **turned out to be erroneous** by the following measurements: (i) the centroid $|\sum z|$ differs between the two floors (make_parent 0.25–1.41 / integer-K floor ≤ 0.24 , with the integer-K floor $\sim 10^{-15}$ for $4 \mid N$). (ii) the $\pm i$ pair structure differs (Paper 4 / Experiment 15: make_parent has no exact $\pm i$ pairs / the integer-K and cyclic-Fourier floors have them for $N = 8k$). Correctly, the two floors are not approximate/exact versions of the same floor but **distinct members** of the relative-equilibrium moduli (the family of self-consistent fixed points selected by the generating measure) satisfying the same ignition qualification (unstable relative equilibrium, 90-degree phase lattice). The individual facts of Paper 3 (the make_parent floor is an approximate relative equilibrium, the integer-K floor is an exact relative equilibrium, the magnitude of the residuals) are correct, but the implication of regarding the two floors as the same floor was erroneous.

Correction 2: the $\pm i$ photon-pair factorization is a property of the high-symmetry theoretical floor (cyclic-Fourier floor) and does not hold exactly for the data floor. Difference: the full $\pm i$ -pair coverage ($N = 8k$) and the p -wave condensate of Paper 4 are properties of the cyclic-Fourier floor $z_{ij} = r_d e^{i2\pi(i+j)/N}$. The make_parent data floor has no exact $\pm i$ pairs (Paper 4 / Experiment 15: strict pairs 0 for all $N \leq 40$, best $\varepsilon = 3.71 \times 10^{-8}$). Correction: read the factorization of capstone §3.5 with the floor family restricted—exact for the high-symmetry theoretical floor, only near pairs for the data floor.

Correction 3: $N = 3$ splits into ignition/neutral by floor bifurcation. Difference: $N = 3$ has two bifurcations, the T-type ($\sigma = \sqrt{2}$, $|\sum z| = 0.707$, ignites; make_parent floor / integer-K floor) and the triplet type ($\mu/r^2 = -3/2$, rank 1, neutral; cyclic-Fourier floor). Correction: the “ $N = 3$ is the $-3/2$ exception” of capstone §3.7 refers to the triplet-type bifurcation. The $N = 3$ that runs in the data series is the T-type and ignites at onset 45 (Paper 2 / Experiment 13).

By the above, the capstone’s claims about the floor are unified to “data floor = make_parent floor (90-degree lattice, unstable relative equilibrium, nonzero centroid, no exact $\pm i$ pairs).” The integer-K analytic floor of Paper 3 is positioned as a distinct member on the same 90-degree skeleton (not an exact version but a different floor of the same qualification), and the cyclic-Fourier floor of Paper 4 as a different high-symmetry relative-equilibrium family not sharing the 90-degree skeleton. The igniting branches of the three share the ignition qualification “an unstable relative equilibrium,” but are distinguished by the floor’s skeleton, centroid, and $\pm i$ pair structure. However, the cyclic-Fourier triplet type of $N = 3$ ($\mu/r^2 = -3/2$, rank 1) is a neutral branch and does not ignite (Correction 3).

References

1. Noriaki Kihara, “Mechanism of inflationary rapid expansion in self-consistent relational-wave closed systems,” Version DOI 10.5281/zenodo.22176949 / Concept DOI 10.5281/zenodo.22112008. (The source for the definition of the dynamics of this paper, the kinematization of zero-closure, and $\tau \neq$ time.)
2. This series, Papers 1–9 (Foundations of Self-Consistent Relational-Wave Closed Systems, Concept DOIs in Appendix A).
3. Y. Kuramoto, *Chemical Oscillations, Waves, and Turbulence*, Springer (1984); “Self-entrainment of a population of coupled non-linear oscillators”, 1975.
4. L. D. Landau, “On the theory of phase transitions”, 1937; J. Goldstone, *Nuovo Cimento* 19 (1961) 154; P. W. Anderson, *Phys. Rev.* 130 (1963) 439; P. W. Higgs, *Phys. Rev. Lett.* 13 (1964) 508.
5. L. N. Trefethen and M. Embree, *Spectra and Pseudospectra*, Princeton (2005).
6. D. Cvetković, P. Rowlinson, S. Simić, *An Introduction to the Theory of Graph Spectra*, Cambridge (2010).
7. J. E. Marsden and T. S. Ratiu, *Introduction to Mechanics and Symmetry*, Springer (1999).

Appendix A: List of self-citation papers

Because of the volume, this is consolidated in an appendix rather than the main text.

(a) Foundational papers of this series (Papers 1–9; the sources for the details, exhaustive data, and figures of each section of this capstone)

Paper	Title (abbreviated)	Corresponding section	Version/Concept DOI
Paper 1	Single-wave equation of motion; condition for the self-consistent floor on the 90-degree skeleton (amplitude eigenvalue problem)	§2, §4	Version 10.5281/zenodo.22728761 / Concept 10.5281/zenodo.22728760

Paper	Title (abbreviated)	Corresponding section	Version/Concept DOI
Paper 2	Qualification screening of the initial value = unstable relative equilibrium	§3.6	Version 10.5281/zenodo.22728824 / Concept 10.5281/zenodo.22728823
Paper 3	Analytic exact solution of the 90-degree floor (integer-K $\sigma_{\max}$, parity bipartite, arbitrary-N closed form, $O(N)$)	§3.3	Version 10.5281/zenodo.22728996 / Concept 10.5281/zenodo.22728995
Paper 4	Zero-closure factorization theorem ($L=N/\gcd(N,4)$, $p=\text{spf}(L)$, 2-adic selection rule)	§3.5	Version 10.5281/zenodo.22729081 / Concept 10.5281/zenodo.22729080
Paper 5	Spontaneous symmetry breaking and the degenerate vacuum (all $N \geq 40$, $175/175$, degenerate family not a finite discrete orbit)	§5	Version 10.5281/zenodo.22729089 / Concept 10.5281/zenodo.22729088
Paper 6	Inflation is the linear instability of the floor (finite-time Lyapunov rate of the tangent cocycle)	§6	Version 10.5281/zenodo.22729096 / Concept 10.5281/zenodo.22729095
Paper 7	Geometry after locking (Grassmann plane switching, 4-dimensional span, 2 nonzero principal angles, $SO(4)$ rejected)	§7	Version 10.5281/zenodo.22729108 / Concept 10.5281/zenodo.22729107
Paper 8	Additivity law $a \propto \sqrt{H/M}$ (intensive state quantity, selective erasure of memory)	§8	Version 10.5281/zenodo.22729116 / Concept 10.5281/zenodo.22729115
Paper 9	Anonymity and the minimal unit of information (study paper, $C \propto$ $\simeq Q_{\{k-2\}}/S_k$)	§8	Version 10.5281/zenodo.22729128 / Concept 10.5281/zenodo.22729127

(b) Already-published parent papers (the underlay of this series)

Paper	Version DOI	Concept DOI
Mechanism of inflationary rapid expansion in self-consistent relational-wave closed systems (v2)	10.5281/zenodo.22176949	10.5281/zenodo.22112008
Onset-mode discrimination (amplification $\iff$ unstable relative equilibrium, all 86 states)	—	10.5281/zenodo.21798854
Thermodynamic readout from a self-consistent complex relational system without assuming spacetime (Λ , cosmological term)	10.5281/zenodo.22240034	10.5281/zenodo.22240033
Readout of spacetime and wave metric from anonymous complex zero-closure (log spacetime, $c=1$)	10.5281/zenodo.22282218	10.5281/zenodo.22282217
Conditions under which a seed generates particles and the lower bound of resolution (Paper A)	10.5281/zenodo.21874482	10.5281/zenodo.21874481
Reproduction specification (capstone + 3 chapters)	10.5281/zenodo.22317636	(to be confirmed)

(c) Not yet published but planned for mutual citation (to be cross-cited upon publication); in-series experiments

- **Precision / seed-isolation experiment (100 digits)** `N7_N8_DeltaTau_denominator_wide_sweep_20260902/precision` (the source for the seed-floor verification of §3.6. At IC100, closure $\sim 10^{-102}$, floor $f_{\perp} \sim 10^{-200}$, γ_{τ} agrees to 8 digits with IC64, grows about 179 digits from the floor by a single exponential law and fires. README / ANALYSIS / results/precision_summary.csv). Related: multiprecision 50-digit `Nall_linear1000000_steps5_mpmath50_equimodular_selfconsistent_directHperp_20260828/`.
- **Reproduction A: ramp excess = time-varying non-normal product (direct Lyapunov)** `位相ロック全 N 確認_相対平衡_20260912/ランプ直接 Lyapunov_非正規時変積_20260912/` (direct confirmation of the non-normal correction mechanism of §6. The $H_{\perp}$ slope of the time-varying product $\prod J(z_t)$ matches the measurement and exceeds the floor freeze by 18–31%, $N = 6, 8, 10$. `analyze_ramp_lyapunov_20260912.py` / `results/ramp_lyapunov_summary.csv` / REPORT / SHA / run_all).
- Periodic table of waves_completed paper_v1 (charge clock / mass clock; planned source for the open clock theory of §9).
- Discovery ledger `発見台帳_干渉保存力学_20260905 起点.md` (the unified record of experiments and studies = the backbone of reproduction).

Appendix B: List of figures, generating programs, and relative paths (reproducibility)

We list the content of each figure, its **generating program**, and its **figure relative path** (relative to this capstone file). The `../` in the paths points to directly under this series (干涉保存力学_資格審査とシード無し系列_20260831/). The dynamics map is identical to the canonical `run_N3_N40_stage123_v1.py` (SHA256 1abf2353...) in all figures, with no physical change.

Figure	Content	Generating program (relative path)	Figure relative path
Figure 1	Single-wave EOM verification (measured vs derived, $y = x$)	<code>../位相ロック全 N 確認 _相対平衡_20260912/単 一波運動方程式_導出と 検証 _20260912/make_figures_eom_90deg_fig_1.png (verification verify_single_wave_eom_20260912.py)</code>	<code>../位相ロック全 N 確認 _相対平衡_20260912/単 一波運動方程式_導出と 検証 _20260912/fig_90deg_fig_1.png</code>
Figure 2	90-degree necessity ($\sin 2\Delta\varphi$ zeros + floor/lock distribution)	<code>same as above make_figures_eom_90deg_相対平衡_20260912/単 (verification verify_90deg_necessity_検証_20260912.py)</code>	<code>../位相ロック全 N 確認 _相対平衡_20260912/単 一波運動方程式_導出と 検証 _20260912/fig_90deg_necessity.png</code>
Figure 2b	Algebraic exact solution of the 90-degree floor (integer-K floor, all-N complex diagram)	<code>../90 度理論床_整数 K 解析解_全 N_20260909/make_90deg_相対平衡_20260912/単 (closed-form verification closed_form_90deg_floor_verify_v1.py, parity parity 二部構造 _解析的証明 _20260912/verify_parity_bipartite_analytic_20260912.py)</code>	<code>../90 度理論床_整数 K 解析解_全 _20260912/fig_90deg_floor_complex.png</code>
Figure 3	Mexican-hat effective potential (N=6 / N=40)	<code>../位相ロック全 N 確認 _相対平衡_20260912/自 発的対称性の破れ_シード依存縮退真空 _20260912/make_figure_mexican_hat_20260912.png (N40 is</code>	<code>../位相ロック全 N 確認 _相対平衡_20260912/自 発的対称性の破れ_シード依存縮退真空 _20260912/fig_90deg_fig_3.png (N40: make_figure_mexican_hat_20260912.py)sb_N40.png)</code>
Figure 4	SSB degenerate vacua ($N = 6$, real data)	<code>../位相ロック全 N 確認 _相対平衡_20260912/自 発的対称性の破れ_シード依存縮退真空 _20260912/make_figure_ssb_20260912.py (verification verify_ssb_seed_vacua_20260912.py)</code>	<code>../位相ロック全 N 確認 _相対平衡_20260912/自 発的対称性の破れ_シード依存縮退真空 _20260912/fig_90deg_fig_4.png</code>

Figure	Content	Generating program (relative path)	Figure relative path
Figure 5	SSB for all N 175/175 + $\Delta = -2\pi/N$	../位相ロック全 N 確認 _相対平衡_20260912/自 発的対称性の破れ_シー ド依存縮退真空 _20260912/全 N6_40_SSB バッチ _20260912/make_figure_batch_allN.py (run run_ssb_batch_N6_N40_20260912.py)	../位相ロック全 N 確認 _相対平衡_20260912/自 発的対称性の破れ_シー ド依存縮退真空 _20260912/全 N6_40_SSB バッチ _20260912/make_figure_batch_allN.png
Figure 5a	All- N inflation $H_{\perp}/H$ (step500, same arrangement as complex diagram)	../make_parent 型初期 値_自己無撞着構造 _20260907/wrapper_run_makeparent_v1.py (run) + postprocess_makeparent_sweep_v1.py (plotting)	../make_parent 型初期 値_自己無撞着構造 _20260907/wrapper_run_makeparent_v1.py _20260907/folder_N3vN40/sweep/fig_Hp (run) + postprocess_makeparent_sweep_v1.py (plotting)
Figure 5b	Terminal complex diagram (make_parent floor step500, all N , equal-amplitude ring)	../make_parent 型初期 値_自己無撞着構造 _20260907/wrapper_run_makeparent_v1.py (run) + plot_complex_planes_makeparent_v1.py (plotting)	../make_parent 型初期 値_自己無撞着構造 _20260907/wrapper_run_makeparent_v1.py _20260907/folder_N3vN40/sweep/fig_co (run) + plot_complex_planes_makeparent_v1.py (plotting)
Figure 5c	Inflation $H_{\perp}/H$ ($N = 22$, 2000step)	../位相ロック全 N 確認 _相対平衡 _20260912/N22_2000step 検証走行 _20260912/plot_inflation_N22_2000step (run wrapper_run_N22_2000_v1.py)	../位相ロック全 N 確認 _相対平衡 _20260912/N22_2000step 検証走行 _20260912/plot_inflation_N22_2000step _20260912/fig_Hperp_N22_den22_2000 (run wrapper_run_N22_2000_v1.py)
Figure 5d	Inflation $H_{\perp}/H$ ($N = 40$, 2000step)	../位相ロック全 N 確認 _相対平衡 _20260912/N40_2000step 検証走行 _20260912/plot_inflation_N40_2000step (run wrapper_run_N40_2000_v1.py)	../位相ロック全 N 確認 _相対平衡 _20260912/N40_2000step 検証走行 _20260912/plot_inflation_N40_2000step _20260912/fig_Hperp_N40_den40_2000 (run wrapper_run_N40_2000_v1.py)
Figure 5e	Terminal complex step2000 ($N = 22$, equal-amplitude ring)	../位相ロック全 N 確認 _相対平衡 _20260912/N22_2000step 検証走行 _20260912/plot_complex_plane_step2000 (run)	../位相ロック全 N 確認 _相対平衡 _20260912/N22_2000step 検証走行 _20260912/plot_complex_plane_step2000 (run)

Figure	Content	Generating program (relative path)	Figure relative path
Figure 5f	Terminal complex step2000 ($N = 40$, equal-amplitude ring)	../位相ロック全 N 確認 _相対平衡 _20260912/N40_2000step 検証走行 _20260912/plot_complex_20260912/fig4complexplane_step2000	../位相ロック全 N 確認 _相対平衡 _20260912/N40_2000step 検証走行 _20260912/fig4complexplane_step2000
Figure 6	Inflation = linear instability (N -dependence, ramp, superposition)	../位相ロック全 N 確認 _相対平衡_20260912/イ ンフレ曲線_線形不安定 解析 _20260912/make_figure_inflation_linear_instability (analysis analyze_inflation_linear_instability_20260912.py)	../位相ロック全 N 確認 _相対平衡_20260912/イ ンフレ曲線_線形不安定 解析 _20260912/fig6inflation_linear_instability

Reproducibility upon publication (mandatory): when publishing this capstone, upload in bulk all generating programs of the above table, the input data (floor npz, states npz), the output figures, each package’s README / SHA256SUMS / run_all.sh, and the canonical run_N3_N40_stage123_v1.py, and keep the state in which run_all.sh can regenerate the figures and data. The packages of foundational Papers 1–9 (Appendix A) must likewise be complete.

Peer-review reflection log

- v1→v1.1 (2026-09-12): correction of the coefficient of the §6 onset formula ($\ln(1/H_{\perp 0})/(2 \ln |\mu|) = \ln(1/\text{seed})/\ln |\mu|$); “the unique termwise equilibrium” → “maximally symmetric equilibrium; global uniqueness not claimed”; §5 “vacuum” explicitly stated as a mathematical designation for the degenerate minima of the effective potential; $V(r)$ explicitly noted as an ansatz; §7 separated into the three layers of instantaneous rank 2 / 4D span / double rotation; a caveat on designations added at the opening.
- v1.1→(this series capstone, 2026-09-12): correction of the thinness of the content. With the direction and claims unchanged, §1 was thickened as a research program and causal chain, §3 integrated the qualification screening (Paper 2) and the algebraic exact solution (Paper 3, Figure 2b), §3.4 the factorization (Paper 4), and §8 added the caveat on circular derivation. The details, exhaustive data, and figure sources of each section were linked to foundational Papers 1–9, and Appendix A (list of self-citations) was newly established.
- Content review (2026-09-12): inspected consistency with the foundational papers and stated the three discrepancies in the identification of the floor explicitly in the section “Differences from and corrections to cited papers.” In particular, **the implication of Paper 3 that “the integer- K floor = the exact version of the make_parent floor (the same floor)” was corrected as erroneous by the measurement of the centroid and $\pm i$ pair structure** (the original papers’ description left without weakening, errors stated explicitly as errors). Also removed ambiguous words, excuses, and poorly-grounded numbers (onset “ ≈ 7.4 – $7.9N$ ”, etc.), and explicitly noted unestablished derivations (onset closed form, $V(r)$ first principles, S^3 identification, etc.) as future

work.

- Fixing of the sign convention (2026-09-12, cross-checked against real data / canonical code): the floor is the eigenmode of the **most-negative eigenvalue in absolute value** $\mu = -\sigma_{\max}$ of iK , with $d\varphi/d\tau = -\mu = \sigma_{\max} > 0$ (consistent with the phase equation $r\dot{\varphi} = \sum r_f \sin^2 \Delta\varphi \geq 0$; for $N = 6$ the make_parent floor $\mu = -4.868$ / integer-K floor $\mu = -4.899$, both the minimal eigenvalue of iK). The notation “the maximal eigenvalue of iK ” of §3.3 and Paper 3 was sign-inconsistent and was corrected (the designation $\sigma_{\max} = |\mu|$ = spectral radius = maximal singular value is retained; the form of the floor is $z = \sigma_{\max}u - iKu$). Note that the $|\mu_{\max}|$ of §6 and Paper 6 is the spectral radius of the floor Jacobian J and is a quantity distinct from this μ .
- $SO(4)$ discriminating computation (2026-09-12, real data / canonical code): constructed the cumulative map $M_t = \prod \exp(\Delta\tau K(\varphi_n))$ and inspected the invariance of the 4-dimensional subspace $S = \text{span}(P_0 \cup P_{\text{term}})$. $M_t S = S$ breaks (out-of- S component 0.84–0.99, M_S non-orthogonal $\|M_S^T M_S - I\| = 1.21$, $N = 6$). Hence **“the dynamics is an $SO(4)$ double rotation on four dimensions” is rejected**, and §7 / Paper 7 were corrected to “4-dimensional span, 2 nonzero principal angles, double rotation of the relative configuration (the cumulative map is not $SO(4)$).” The already-established $\dim(P_0 + P_{\text{term}}) = 4$ and 2 nonzero principal angles are retained.
- Reinforcement of the citation for the seed floor (2026-09-12): §3.5 added **an experiment that verified the floor of the seed at 100-digit precision** (precision_seed_dynamics_isolation_100digit_20260902, $N = 7, 8$). It has been shown that even lowering to closure $\sim 10^{-102}$, floor $\sim 10^{-200}$ at IC100, γ_τ agrees to 8 digits with 64-bit and necessarily fires. By this, the previous version’s description “verification at multiprecision is additional research” is erroneous, and was replaced with a citation as **verified** (removing the defensive hedge).
- Re-review of open matters by cross-checking with existing papers (2026-09-12): cross-checked all “open / additional research” items against the discovery ledger and all packages. **Corrected 2 items that had been treated as open despite being already verified**—(i) $N \geq 18$ lock (established by Paper 5 $N = 22, 40 + N=64$ phase-lock verification, §Paper 2), (ii) transition rank-4 S^3 (inspected as transient with no persistent structure by short-window SVD + discriminating computation, Paper 7). **3 items settled by reproduction**—(A) the non-normal correction mechanism = time-varying non-normal product (the slope of the time-varying product matches the measurement and exceeds the floor freeze by 18–31%, asserted, Paper 6), (B) the terminal block decomposition does not stand for the data series (make_parent floor $\rightarrow$ terminus) and the minimal block is intrinsic to the cyclic-Fourier / integer-K high-symmetry floor (rejected, Paper 9), (C) $\sigma = N - 1 \leftrightarrow$ vertex sector does not correspond, being a coincidence of dimension labels (the 3-decomposition and $\sigma \rightarrow N - 1$ retained, rejected, Paper 8). Each reproduction was saved in a dedicated package (py + results + README + REPORT + SHA + run_all) and cited in the relevant paper. Those left genuinely open ($V(r)$ first principles, the closed form of the onset/amplification rate, coordinates/gauge/clock, the amplitude-preserving control dynamics for circular derivation) are explicitly noted as future work. The $N \geq 18$ long-time run of the theoretical floor was judged unnecessary as it is trivial from the $N = 64$ confirmation of the data floor.
- Synchronization accompanying the arbitrary- N analytification of Paper 3 (2026-09-12): confirmed that only the parity $p_e = (i + j) \bmod 2$ of the species $k_e = (i + j) \bmod 4$ makes the W of Paper 1 the bipartite graph $W_{ef} = A_{ef} \mathbf{1}_{p_e \neq p_f}$ (the W of the iterated floor matches the W of species parity at machine precision, $\| \text{difference} \| \leq 3 \times 10^{-27}$; the floor can be constructed in one shot by $k \rightarrow W \rightarrow \text{Perron} \rightarrow z = Dr$ with no iteration, and matches the iterated floor modulo sign gauge).

By this, the closed forms of Paper 3 (σ^2 , amplitude², full spectrum) were promoted to **arbitrary** N (sympy symbolic + numerical $N = 3\text{--}40$) via the reduced Perron problem, and the active dimension $O(N)$ (for $N \geq 6$, $\text{rank}K = 2(N - 1)$, kernel $(N - 1)(N - 4)/2$; $N = 3, 4, 5$ low-dimensional exceptions rank 2 / 1, rank 2 / 4, rank 6 / 4), and the closed forms of the amplitude-class edge assignment (= parity class) and odd- N residual centroid, which were open in Paper 3, were settled. Furthermore, it was established as a general proof that zero-closure $\sum z^2 = 0$ is a consequence of the bipartite eigenmode ($\|x\|^2 = \|y\|^2$, for all nonzero modes not only the Perron one), that the connectivity of the bipartite W holds for any $N \geq 3$ (recovering the Perron applicability condition of Paper 1), and that the sign-gauge family is 2^{M-1} . The Paper 3 citation of §3.3 was updated from “ $N = 3\text{--}20$ sympy” to “arbitrary N ,” and the W irreducibility from “ $N = 3\text{--}40$ confirmed” to “proved for any $N \geq 3$.” Reproduction 90 度理論床_整数K 解析解_全 N_20260909/parity 二部構造_解析的証明_20260912/.

- **Reconstruction of the capstone after full review of 1–9 (2026-09-13):** integrated into the capstone the results of self-reviewing and promoting Papers 1–9 in order. The Abstract and §1 were fully rewritten with the new causal chain (map $\rightarrow$ 90-degree relative equilibrium $\rightarrow$ ignition qualification $\rightarrow$ **inflationary rapid expansion** $\rightarrow$ SSB / plane reorientation $\rightarrow$ terminal universalization $\rightarrow$ relational invariants) + the central proposition + the unifying axes A/B/C (the old “first... second...” redundant development deleted). §3.5 promoted the factorization theorem ($p = \text{spf}(L)$, $p = 2 \iff 8 \mid N \iff \nu_2(N) \geq 3$) to an independent section, and demoted the ignition qualification to §3.6 and the open matters to §3.7 with renaming (“honest open matters” $\rightarrow$ “established scope and open matters”). §5’s “unstable maximally symmetric equilibrium” $\rightarrow$ “unstable relative equilibrium of the 90-degree skeleton (make_parent),” “35/35” $\rightarrow$ “175/175,” and reflected the vacuum manifold = continuous moduli (720-vertex-permutation test). §6 was changed from “floor eigenvalue + non-normal correction” to a **two-stage dynamics** (the finite-time Lyapunov rate of the tangent cocycle $\Phi(t, 0) = \prod_t J(z_t)$; $\mu_{\max}$ is a lower bound). §7 was changed to the Grassmann formulation ($\Pi_{t+1} = Q_t \Pi_t Q_t^T$, the identity $H_\perp/H = \frac{1}{4}\|\Pi_t - \Pi_0\|_F^2$, exponential departure on $\text{Gr}(2, M)$), and the title updated. A new §8 “Terminal universality, selective erasure of memory, and anonymity” was added (Papers 8 and 9 moved from auxiliary to body, Axes B/C). The old §8 “What is not claimed” was renamed to §9 “Established scope and open problems,” with the 128 registers of Paper A explicitly demoted as scaffolding, and §9 related work $\rightarrow$ §10. The audit criterion was unified to “what is proved was proved / the exhaustive was confirmed / the rejected was rejected / only the underived is open” (removal of evasive language, **not weakening the phenomenon name of inflation by replacing it with a mechanism word**). Also added, from existing figures, the all- N inflation figure (Figure 5a, step500 grid) and the 2000-step inflation + terminal complex for $N=22/40$ (Figures 5c–5f). Appendix B was expanded into a reproducibility table of “figures, generating programs, relative paths” and the bulk-upload policy upon publication was stated.
- **Refinement review of the capstone (2026-09-13, 7 points. Separate theorem from measurement / restrict the subject):** ① restricted the subject of the factorization of §3.5 / the Abstract to “on the cyclic-Fourier high-symmetry floor” (not the data floor in general). ② the reason for the failure of ignition in §3.6 from “was not a relative equilibrium” $\rightarrow$ “was **not an unstable relative equilibrium**” (including the neutral fixed point of (A)). ③ refined the SSB vacuum manifold’s “continuous moduli” to “**not identifiable as a finite discrete symmetry orbit** (S_N + **global phase** + **conjugation**),” explicitly noting the continuity as open by a

seed 1-parameter scan, and deleted “ G/H analytic identification is generally impossible” (for a continuous group, G/H is a continuous orbit). The continuity of the C_k abstract closure space (Q_{k-2}) is retained as it is proved. ④ separated §6’s “ $\mu_{\max}$ is a lower bound of the real rate” into “ $\lambda_0 = \ln \rho_0$ (initial/frozen rate) and the real rate γ_t are distinct quantities; for $N = 6, 8, 10$, $\gamma_t > \lambda_0$; no general proof.” Unified the Jacobian-side notation to $\rho_0 = \rho(J_0), \lambda_0$ (resolving the clash with the μ of iK in the text). ⑤ stated §7’s identity $H_{\perp}(t)/H = \frac{1}{4}\|\Pi_t - \Pi_0\|_F^2$ at all times and strengthened $\|\Pi_t - \Pi_0\|_F \propto e^{\gamma_t t}$ as derived. ⑥ §9 “maximally symmetric single-coherent mode” $\rightarrow$ “self-consistent single-coherent relative equilibrium of the 90-degree skeleton,” and §10 Trefethen–Embree updated to the time-varying tangent-cocycle criterion. ⑦ synchronized old section-number references (factorization §3.4 $\rightarrow$ §3.5, $N = 3$ exception §3.6 $\rightarrow$ §3.7, 100-digit appendix §3.5 $\rightarrow$ §3.6). The title was updated to the current position (local instability $\rightarrow$ inflation $\rightarrow$ SSB $\rightarrow$ Grassmann $\rightarrow$ terminal universality $\rightarrow$ relationalism).

- **Whole-consistency audit just before commit (2026-09-13, 2 substantive points + 1 notational point):** ① revived the $N = 3, 4, 5$ exceptions in the rank formula of §3.3 / the review log—the active dimension $O(N)$ holds in general, but the closed form rank $K = 2(N - 1)$, kernel $(N - 1)(N - 4)/2$ is limited to $N \geq 6$ (explicitly noting the low-dimensional exceptions $N = 3$: rank 2 / kernel 1, $N = 4$: rank 2 / kernel 4, $N = 5$: rank 6 / kernel 4). ② corrected the places that had mixed the cyclic-Fourier floor as “another member of the 90-degree floor” into the three-family separation—(i) make_parent floor (90-degree skeleton), (ii) integer-K analytic floor (the same 90-degree skeleton), (iii) cyclic-Fourier floor (a different high-symmetry relative-equilibrium family, generally not a 90-degree lattice but cyclic symmetry $z_{ij} = r_d e^{i2\pi(i+j)/N}$). Made the three families explicit at the opening and end of “Differences from and corrections to cited papers,” and restricted the subject of the block-size quantization of §1(c) / §8 to “the discrete symmetry of the cyclic-Fourier high-symmetry floor.” The three share the ignition qualification “an unstable relative equilibrium” but are distinguished by the floor skeleton, centroid, and $\pm i$ pair structure. ③ unified the coefficient of the body of $V(r)$ in §5.1 to $\lambda \rightarrow \lambda_0$ ($\lambda_0 = \ln \rho_0$, resolving the notational clash between the parameter definition and the body).

Reproduction

The verification scripts, exhaustive data, and additional figures for each claim of this capstone are stored in a set (py + REPORT + SHA256SUMS + run_all.sh) in the packages of the corresponding foundational papers (Papers 1–9) (relative paths in Appendix A / Appendix B). The dynamics map is identical to the prior paper (Reference 1) (canonical run_N3_N40_stage123_v1.py, SHA256 1abf2353fee2e4f56f05e7a6f149fd086885136beb61ab571b48a56b09691567 cross-checked, physical equations unchanged). The series table of contents is 関係波閉鎖系基礎シリーズ_論文一覧と元資料_20260912.md.